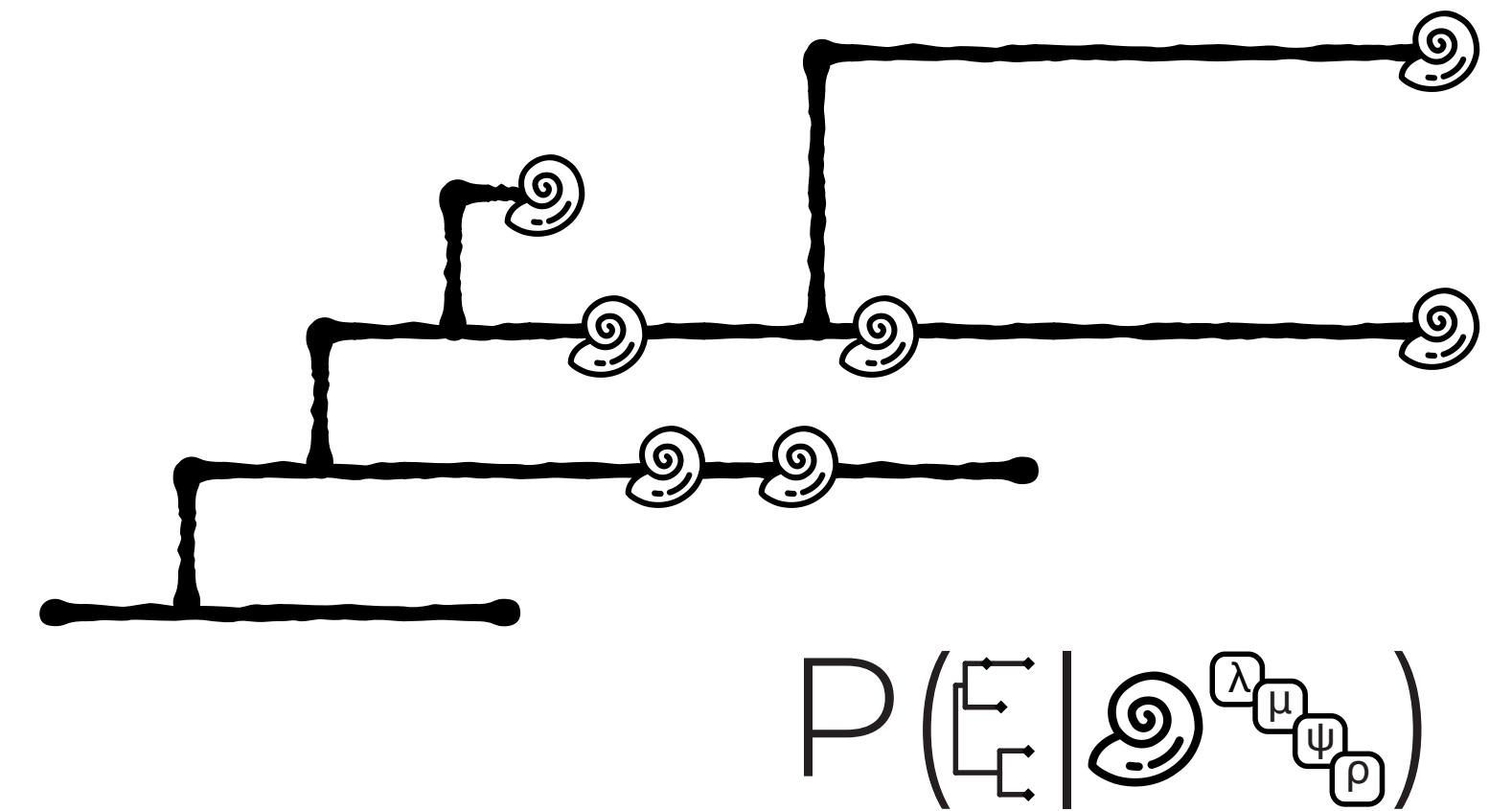
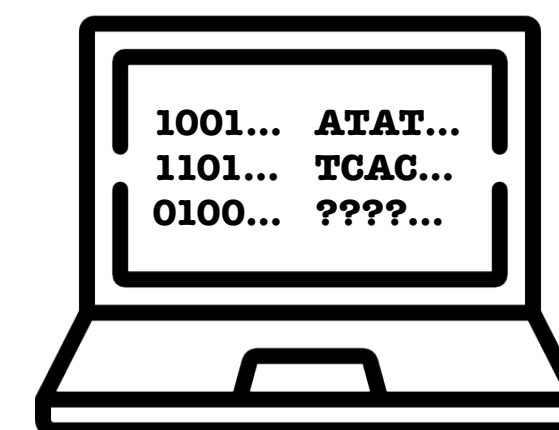
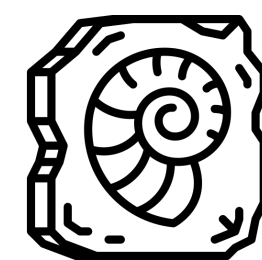


Estimating divergence times using the fossilised birth-death process

Taming the BEAST Beijing

Rachel Warnock

15.07.2025



Objectives

Lecture

- Potential issues with node dating (in brief)
- Dating with sampling through time (the fossilised birth-death process)
- Total-evidence dating

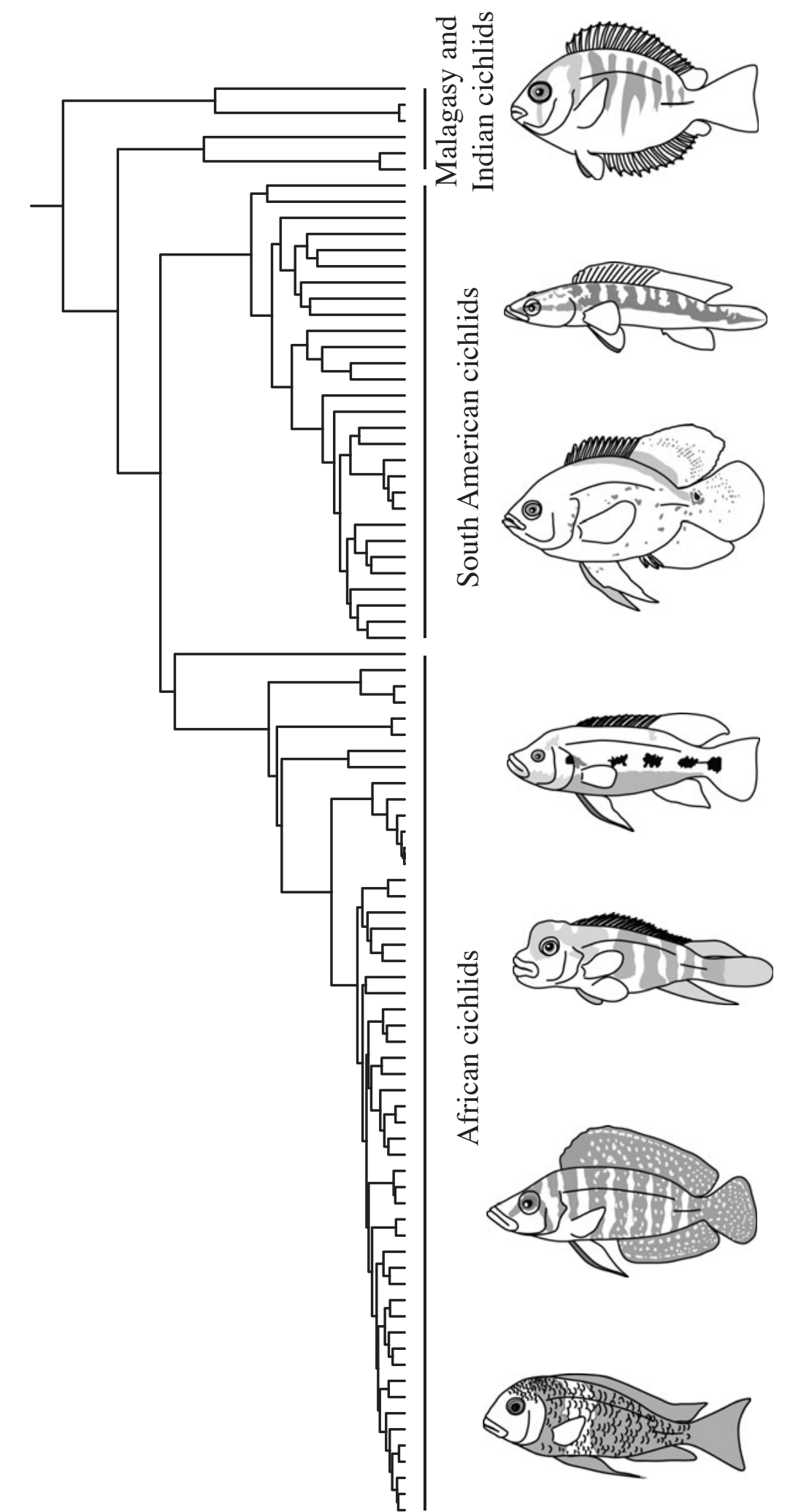
Tutorial

- Divergence dating under the fossilised birth-death process



What can we learn from trees?

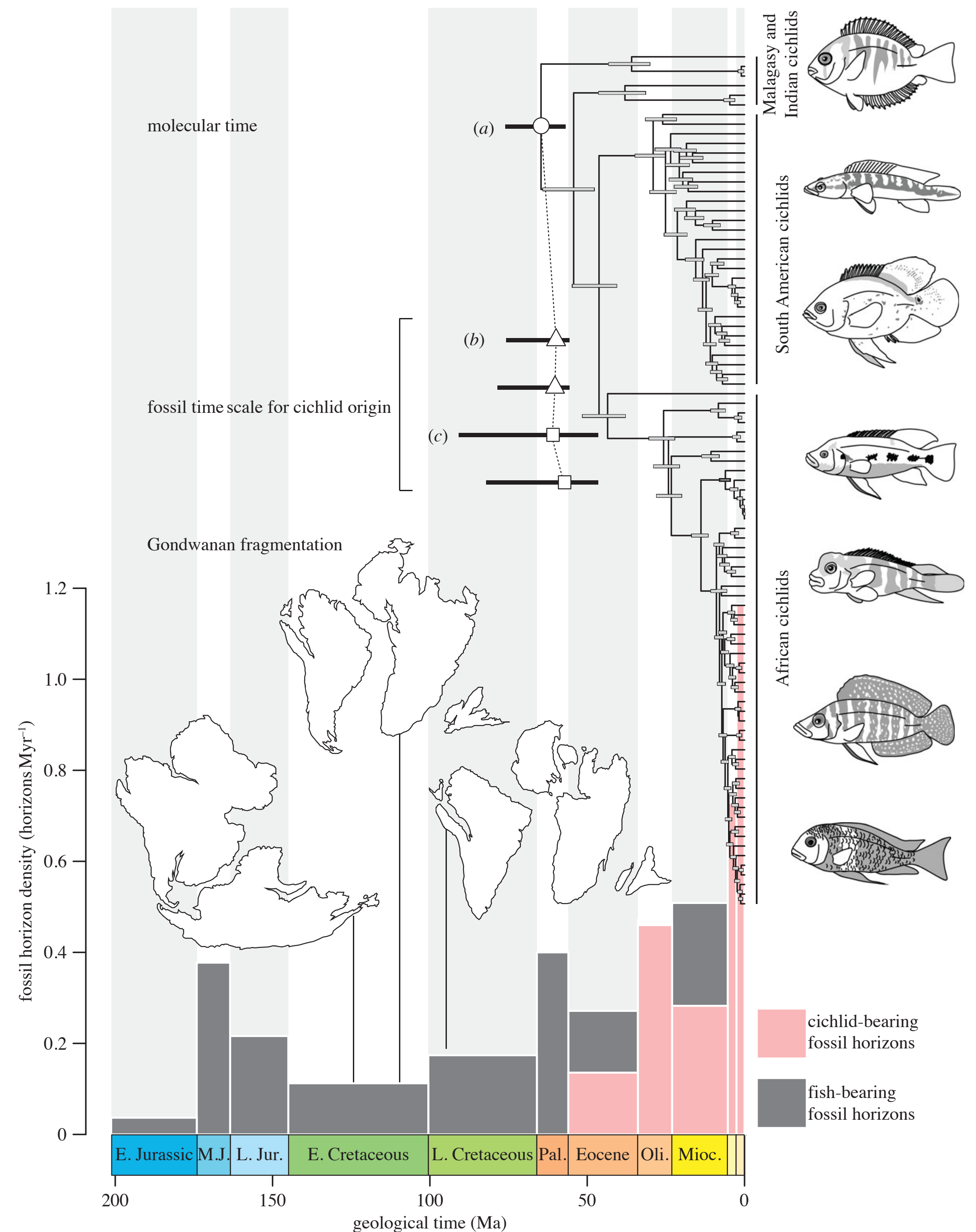
- Evolutionary relationships



What can we learn from trees?

- Evolutionary relationships
- Timing of diversification events
- Geological context
- Rates of phenotypic evolution
- Diversification rates

Image adapted from *Friedmann et al.* ([2013](#))



We use a Bayesian framework

$$P(\text{model} \mid \text{data}) = \frac{P(\text{data} \mid \text{model}) P(\text{model})}{P(\text{data})}$$

Diagram illustrating the Bayesian framework equation:

- likelihood** points to $P(\text{data} \mid \text{model})$
- priors** points to $P(\text{model})$
- posterior** points to $P(\text{model} \mid \text{data})$
- marginal probability of the data** points to $P(\text{data})$

Bayesian divergence time estimation

The data

and / or

0101... ATTG...
1101... TTGC...
0100... ATTC...



phylogenetics
characters

sample
ages

Bayesian divergence time estimation



The data

and / or

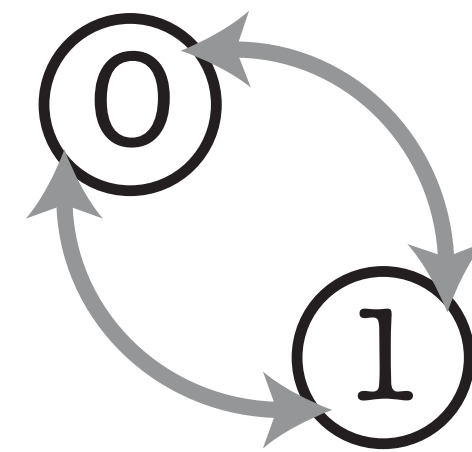
0101... ATTG...
1101... TTGC...
0100... ATTC...



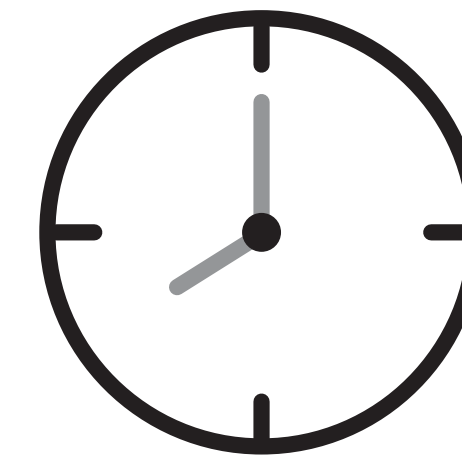
phylogenetics
characters

sample
ages

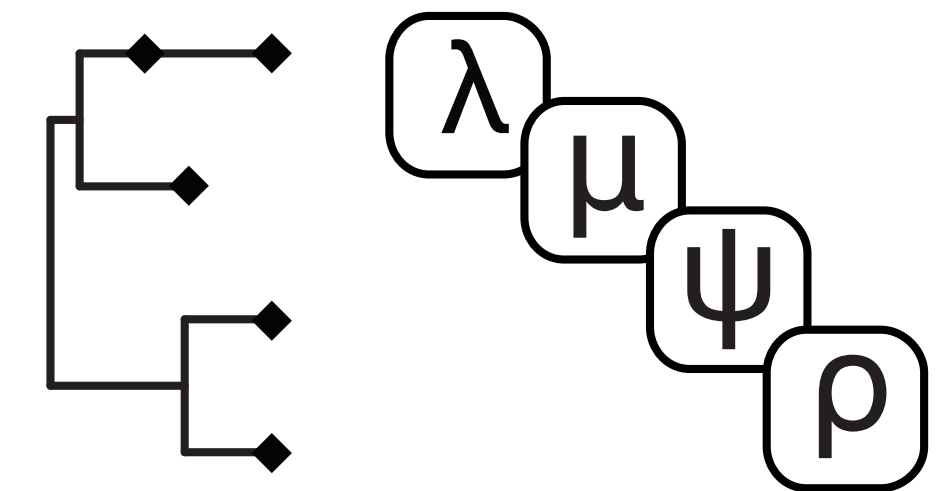
3 model components



substitution
model

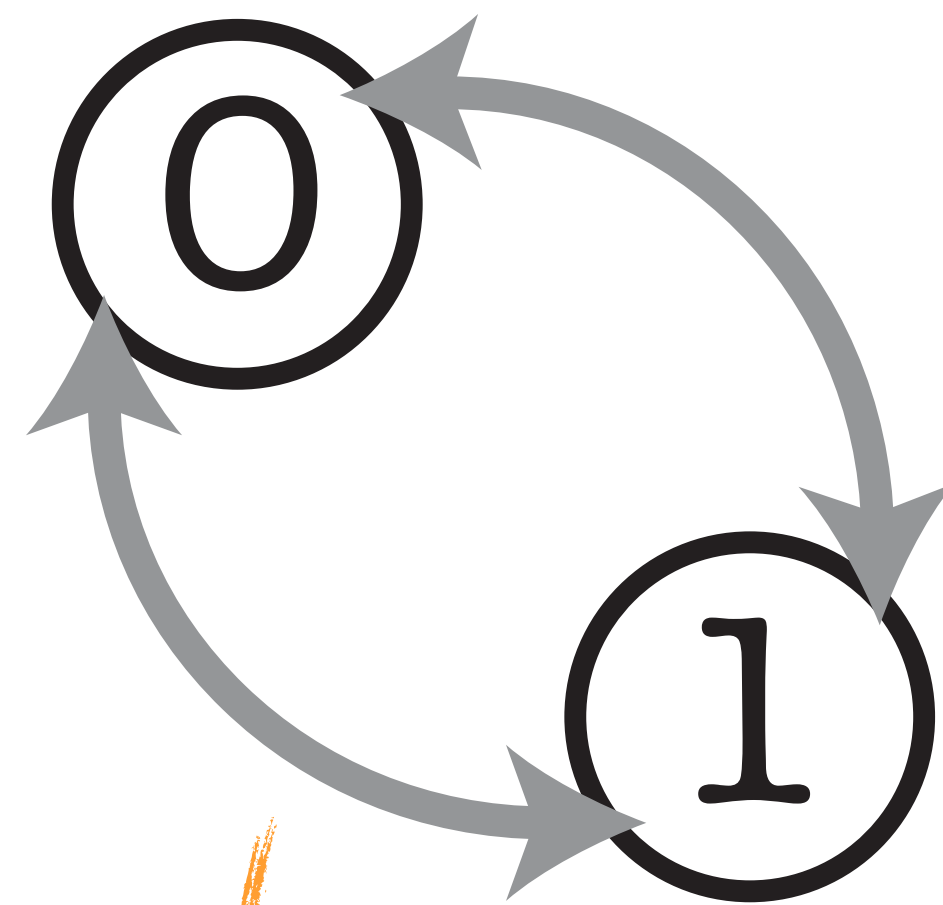


clock
model

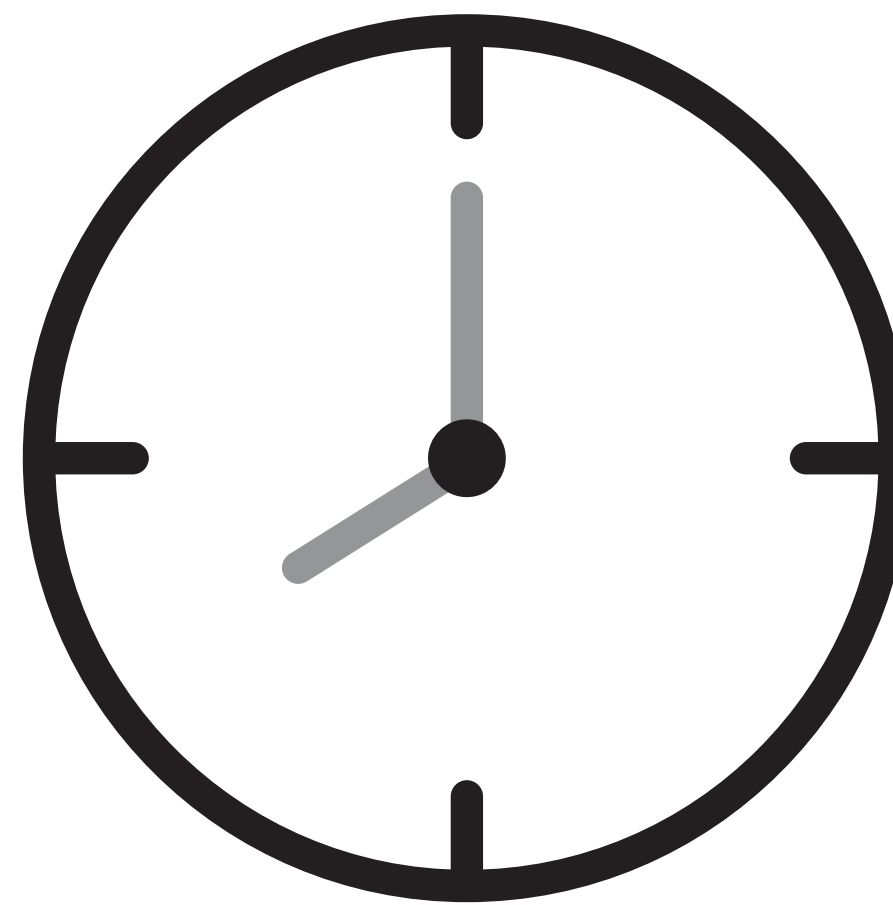


tree and tree
model

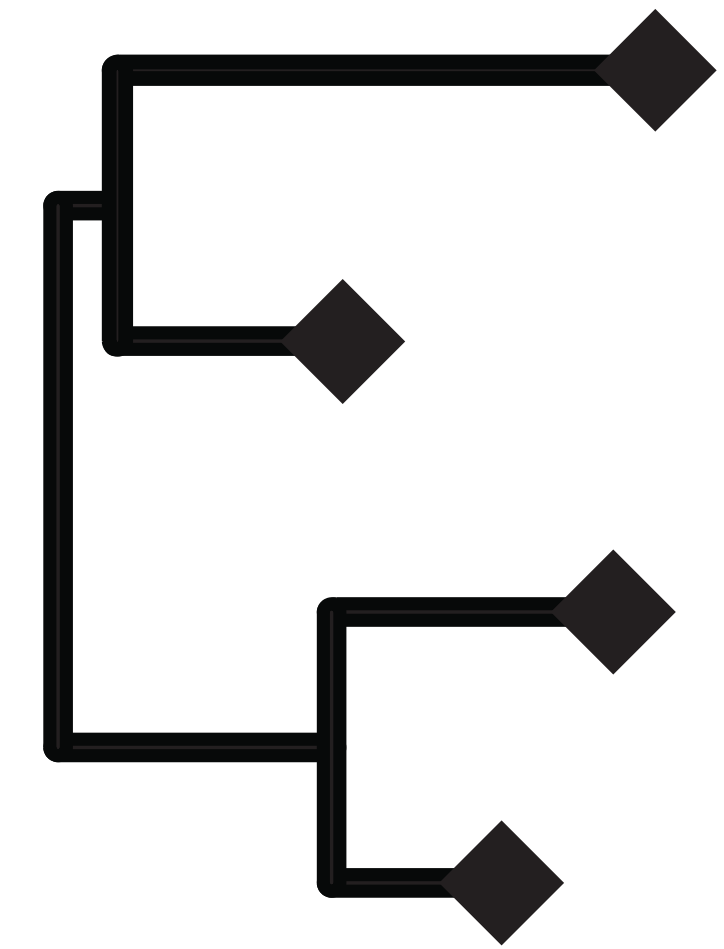
substitution model



clock model

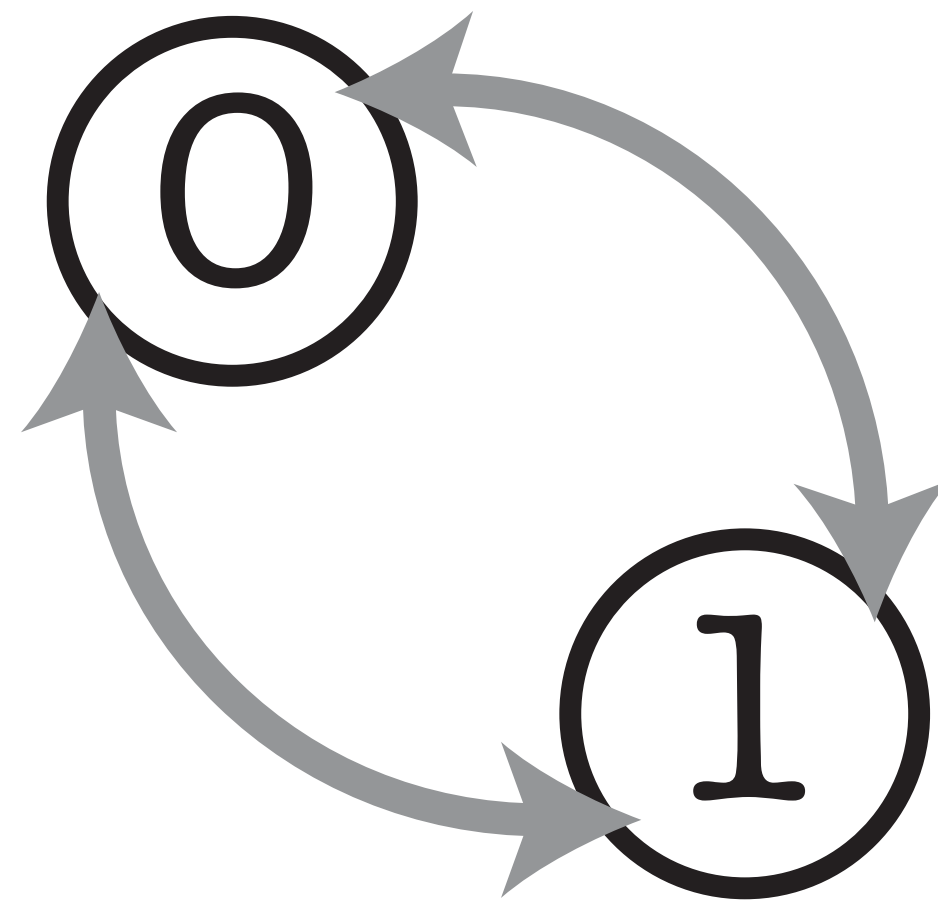


tree model

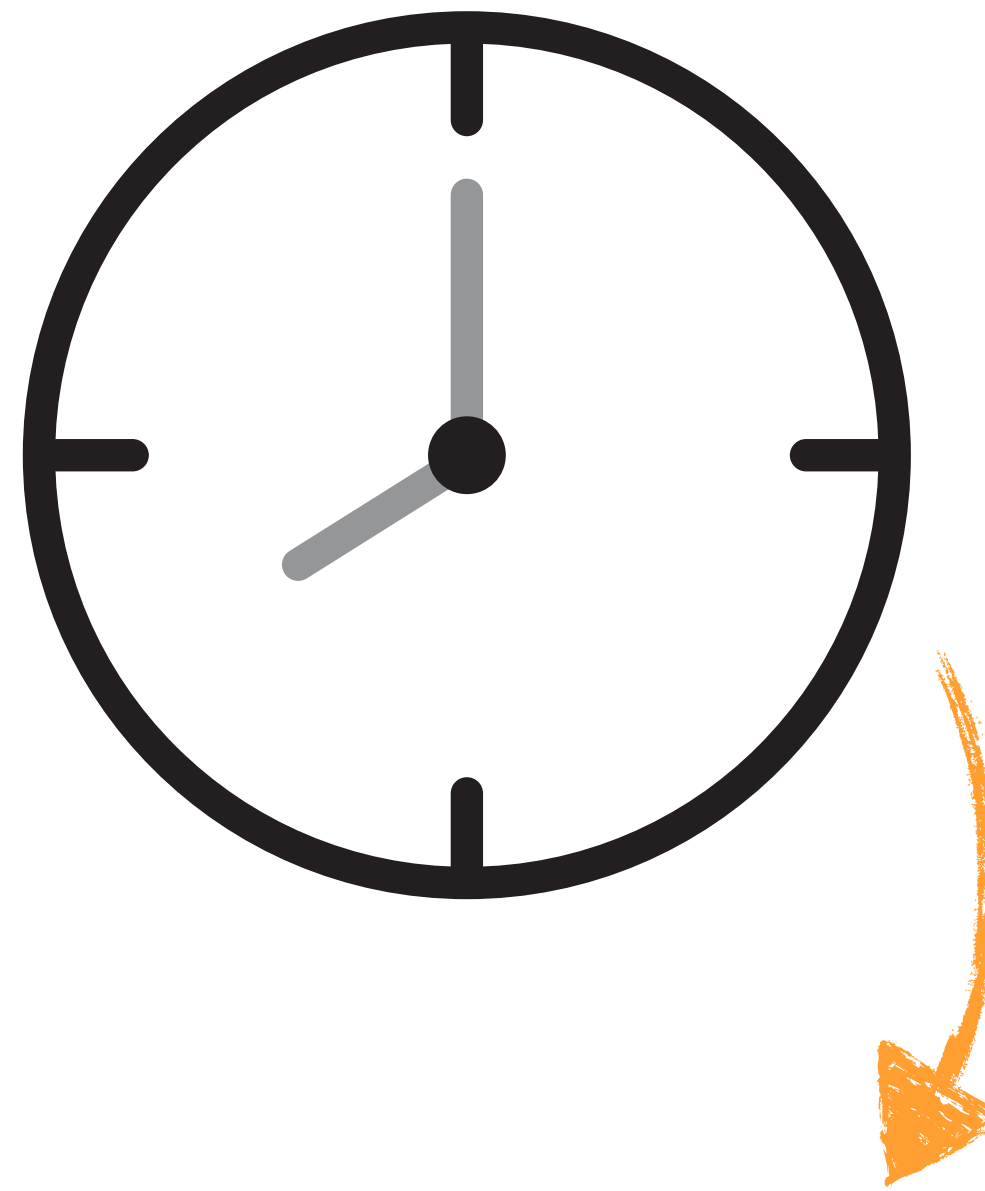


How likely are we to observe a change between character states? e.g., $A \rightarrow T$

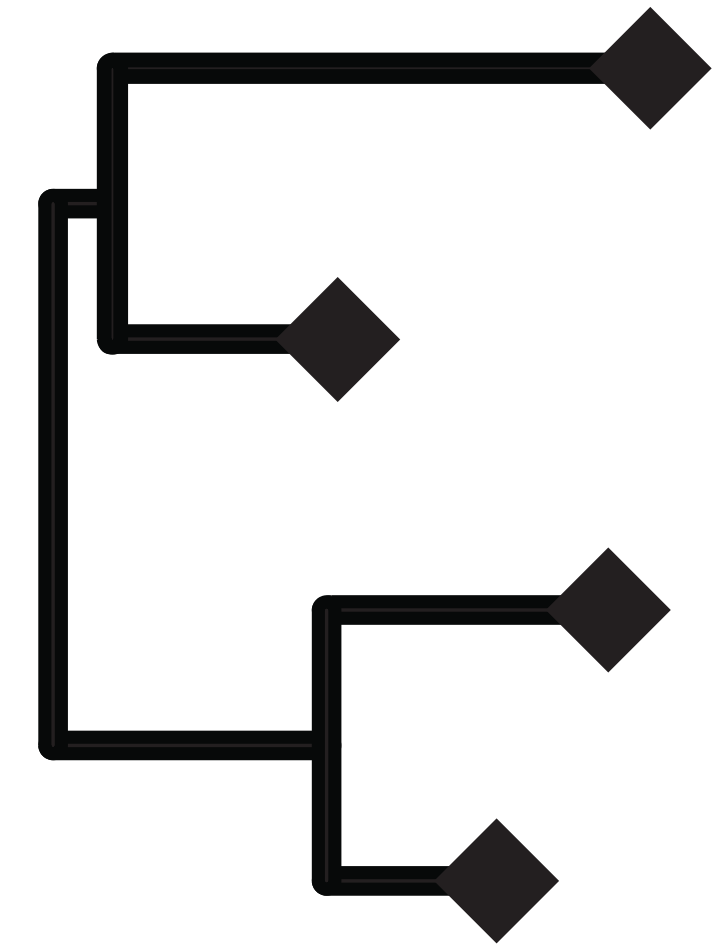
substitution model



clock model

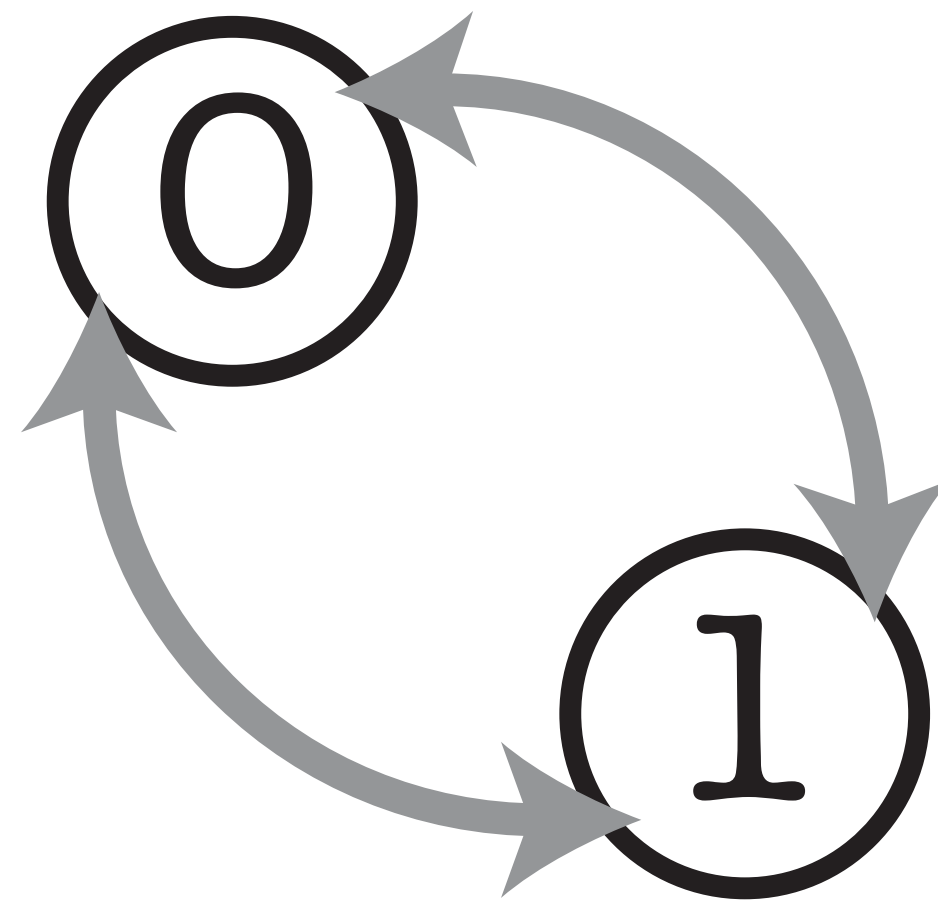


tree model

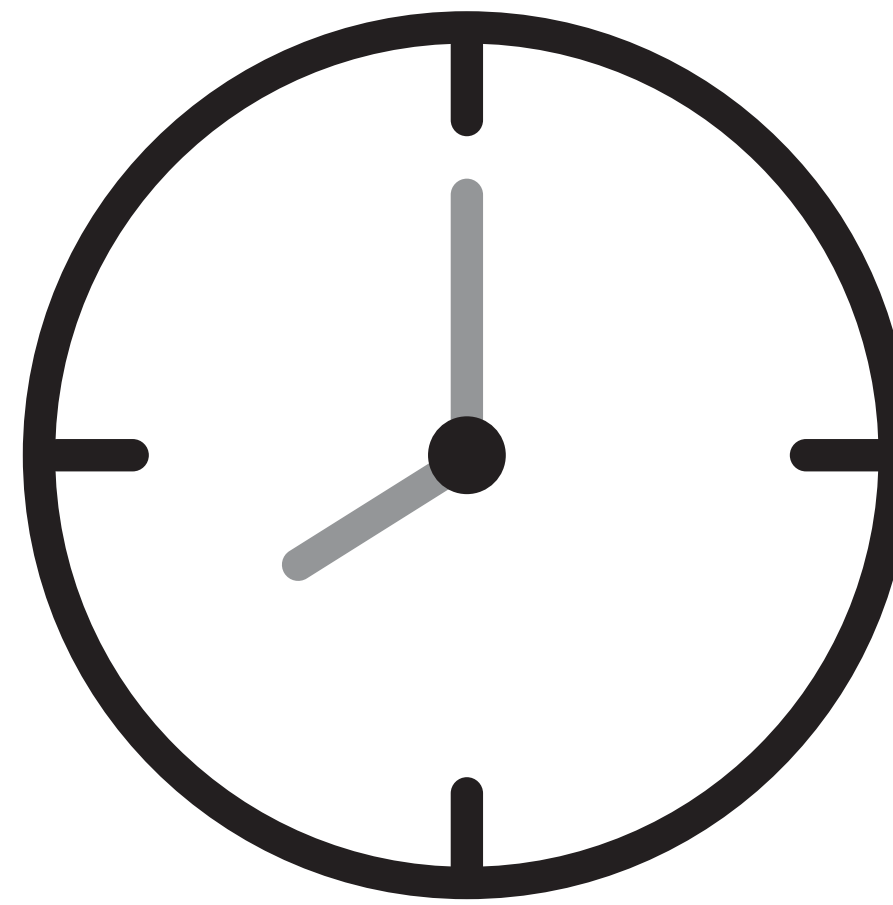


How have rates of evolution varied
(or not) across the tree?

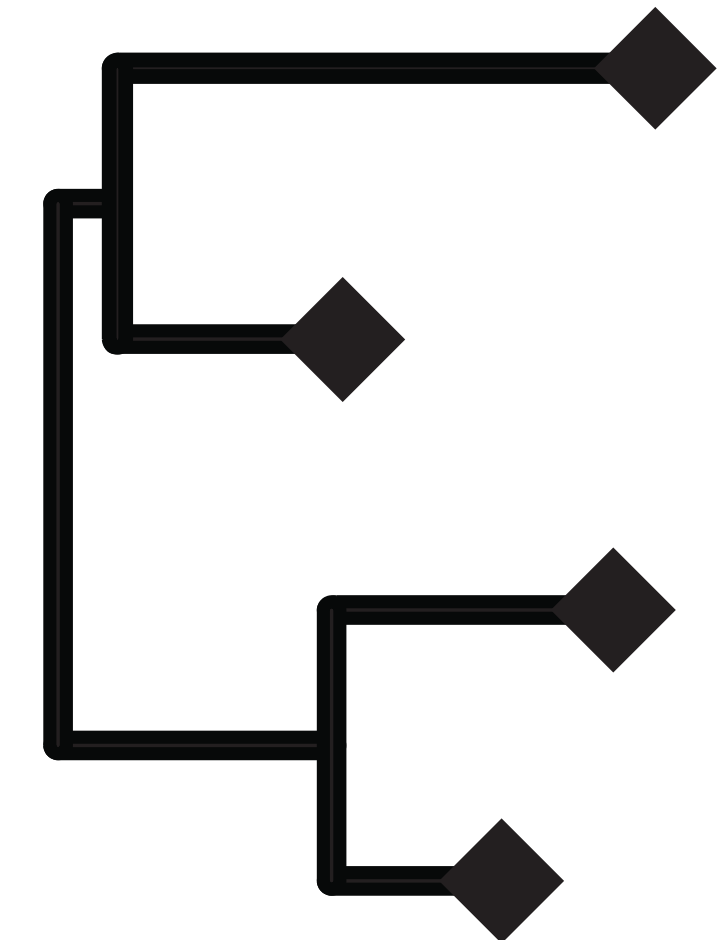
substitution model



clock model



tree model



How have species originated, gone extinct and been sampled through time?

Bayesian divergence time estimation

posterior

$$P\left(\begin{array}{c} \text{tree} \\ \lambda, \mu, \psi, \rho \\ \text{clock} \end{array} \mid \begin{array}{c} 0101\dots \\ 1101\dots \\ 0100\dots \end{array} \text{data}\right) =$$

likelihood

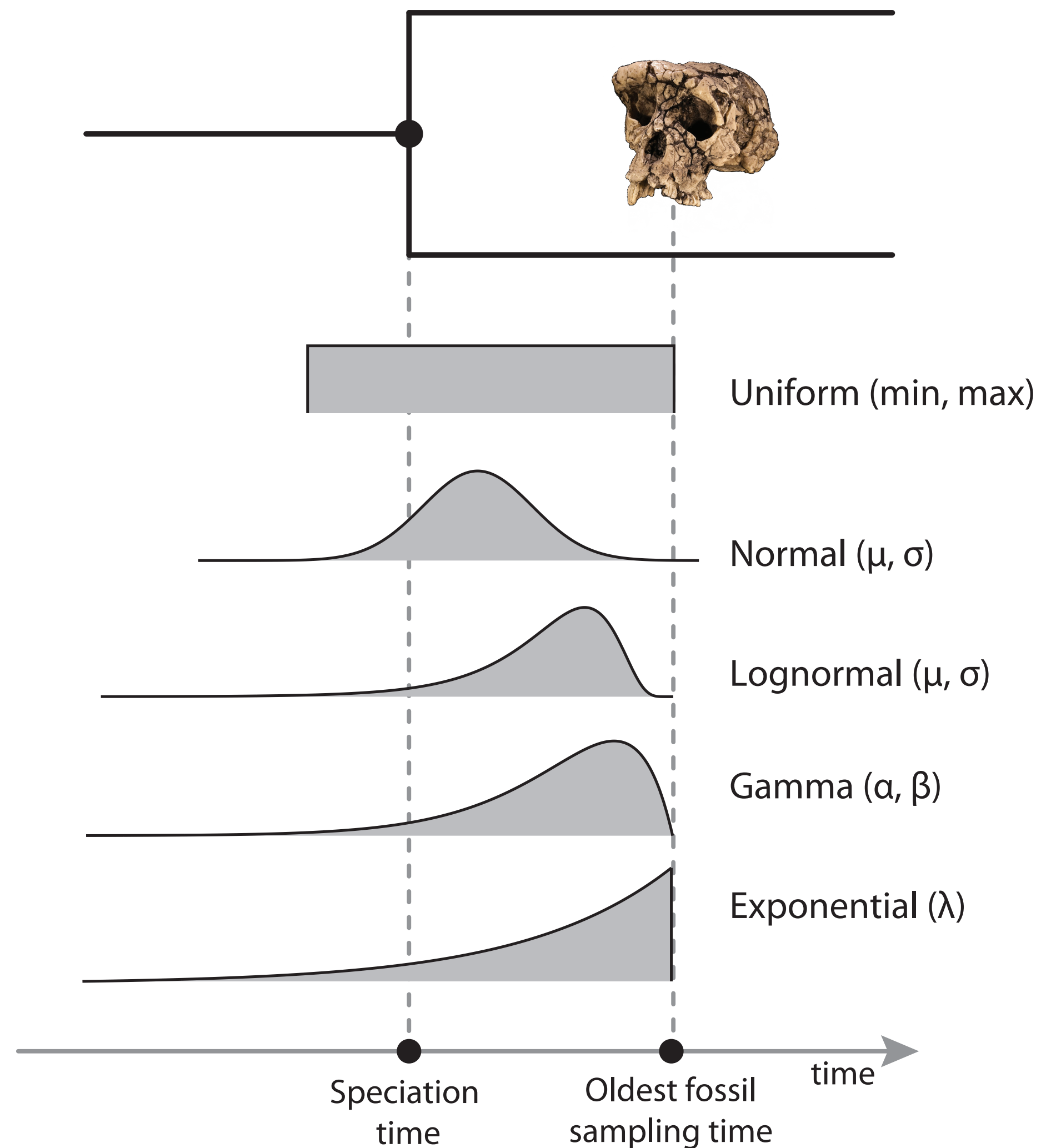
probability of the
time tree

priors

$$\frac{P\left(\begin{array}{c} 0101\dots \\ 1101\dots \\ 0100\dots \end{array} \mid \begin{array}{c} \text{tree} \\ \lambda, \mu, \psi, \rho \\ \text{clock} \end{array}\right) P\left(\begin{array}{c} \text{tree} \\ \lambda, \mu, \psi, \rho \end{array} \mid \text{data}\right) P(\lambda, \mu, \psi, \rho) P(\text{clock})}{P\left(\begin{array}{c} 0101\dots \\ 1101\dots \\ 0100\dots \end{array} \mid \text{data}\right)}$$

marginal pr of the data

Recap: Node dating



We can use a **calibration density** to constrain internal node ages

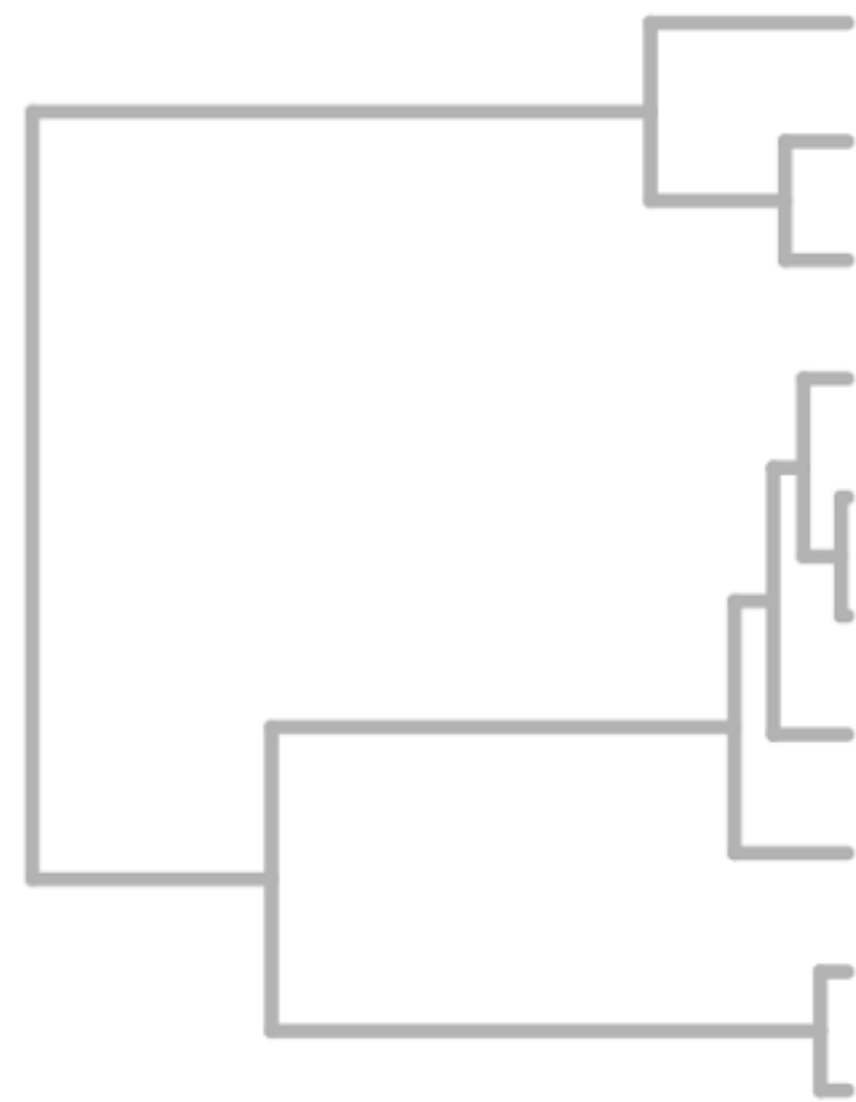
We typically use a **birth-death process** model to describe the tree generating process

Issues with node dating

1. Rate and time is non-identifiable
2. Calibrations are hard to define objectively
3. The “effective priors” do not match the user specified priors
4. We’re potentially excluding a lot of information

The priors will always influence our results

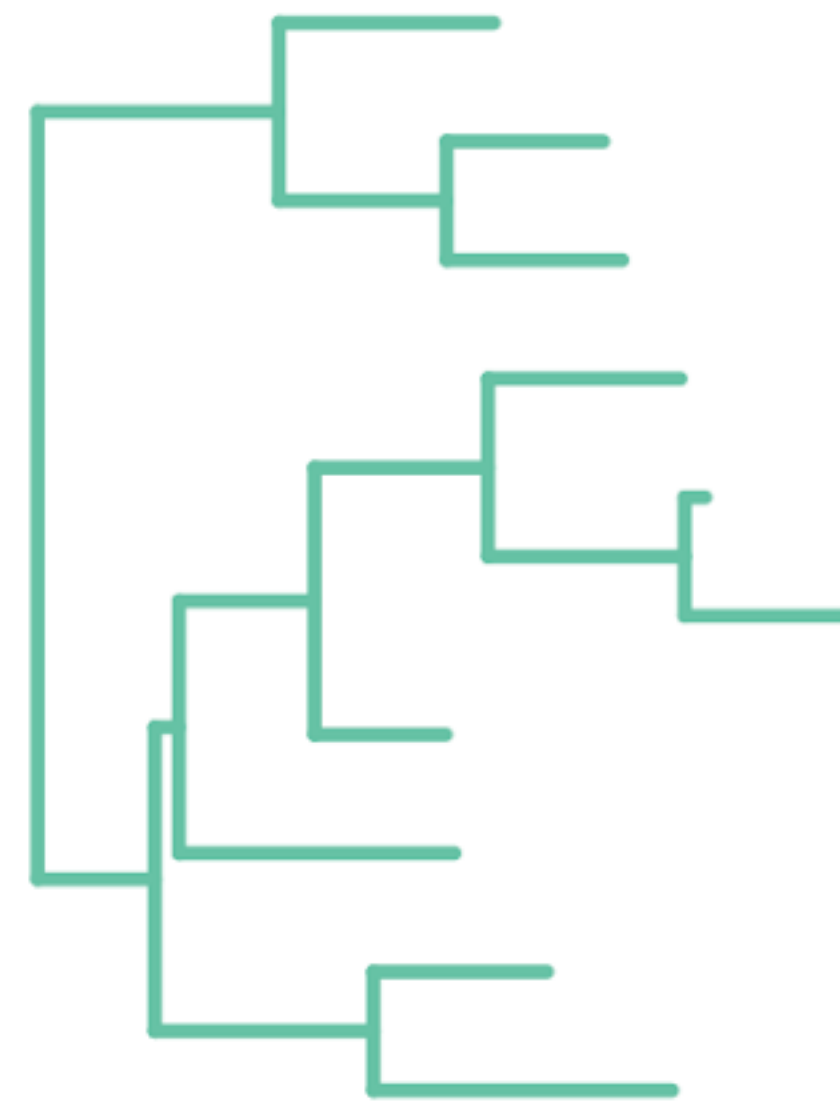
Rate and time are not identifiable - not a conventional Bayesian problem



Time (year)

Prior

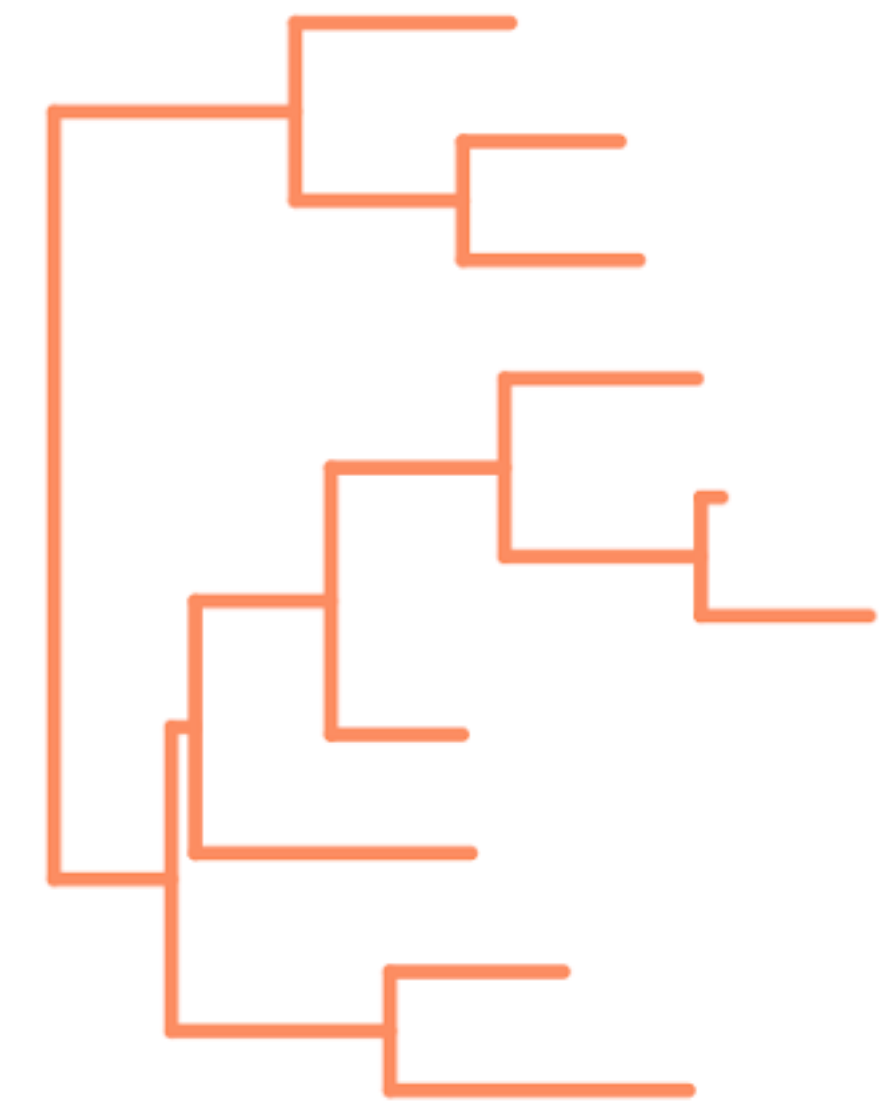
X



Clock rate (subs/site/year)

Prior

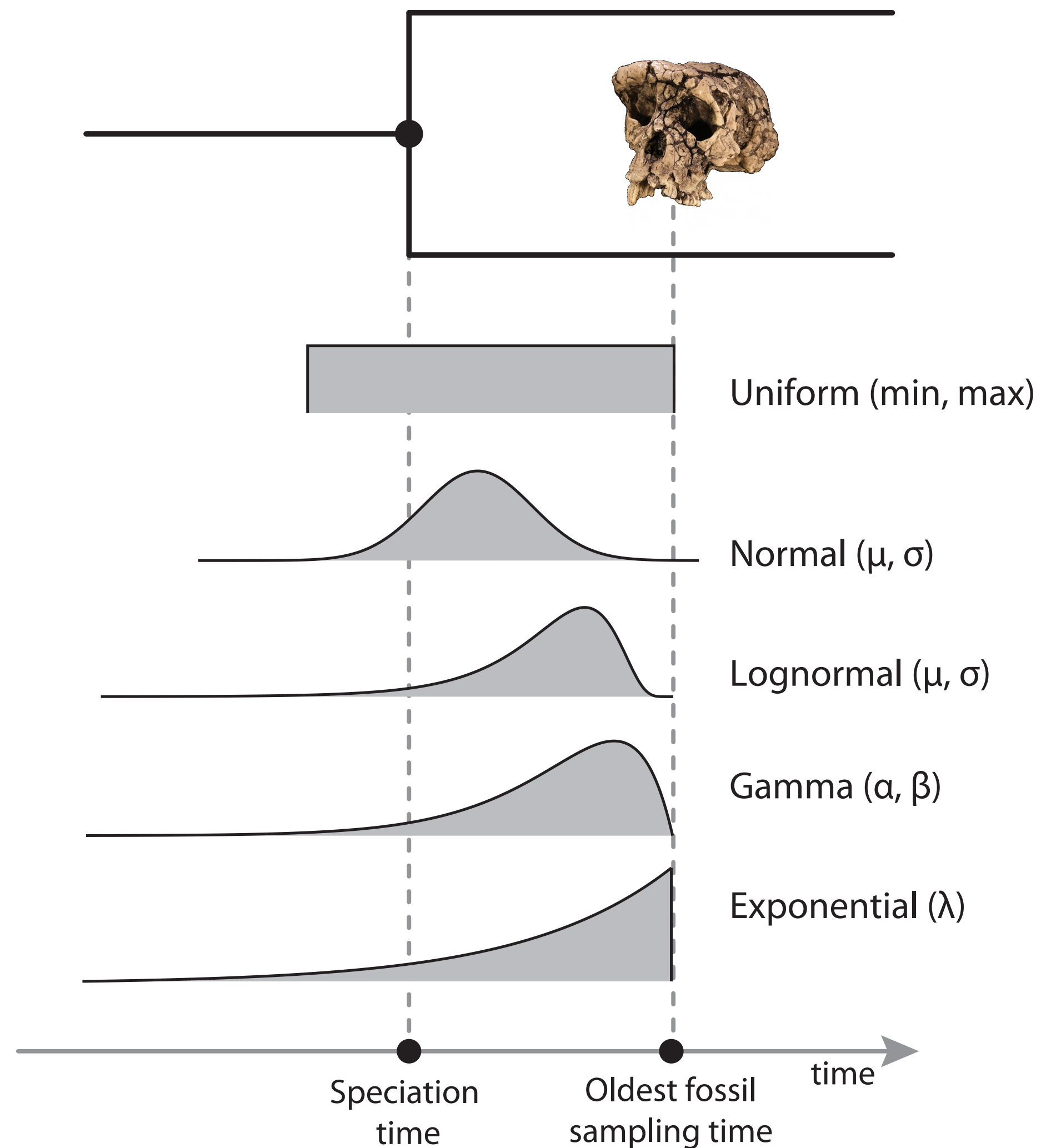
=



Genetic distance (subs/site)

Likelihood

Minimum and maximum constraints



Hard minimum bounds are based on first appearances

Soft maximum bounds are based on more tenuous evidence — typically the 97.5% limit of the calibration density

The upper bounds often influence the results

Maximum constraints

the “no way in hell” argument

Best Practices for Justifying Fossil Calibrations

JAMES F. PARHAM^{1,2,*}, PHILIP C. J. DONOGHUE³, CHRISTOPHER J. BELL⁴, TYLER D. CALWAY⁵,
JASON J. HEAD⁶, PATRICIA A. HOLROYD⁷, JUN G. INOUE⁸, RANDALL B. IRMIS⁹,
WALTER G. JOYCE¹⁰, DANIEL T. KSEPKA^{11,12}, JOSÉ S. L. PATANÉ¹³, NATHAN D. SMITH^{14,15},
JAMES E. TARVER^{3,16}, MARCEL VAN TUINEN¹⁷, ZIHENG YANG¹⁸, KENNETH D. ANGIELCZYK¹⁵,
JENNY M. GREENWOOD³, CHRISTY A. HIPSLEY^{19,20}, LOUIS JACOBS²¹, PETER J. MAKOVICKY¹⁵,
JOHANNES MÜLLER¹⁹, KRISTER T. SMITH²², JESSICA M. THEODOR²³, RACHEL C. M. WARNOCK³,
AND MICHAEL J. BENTON³

Goal - to make calibration choices transparent and explicit

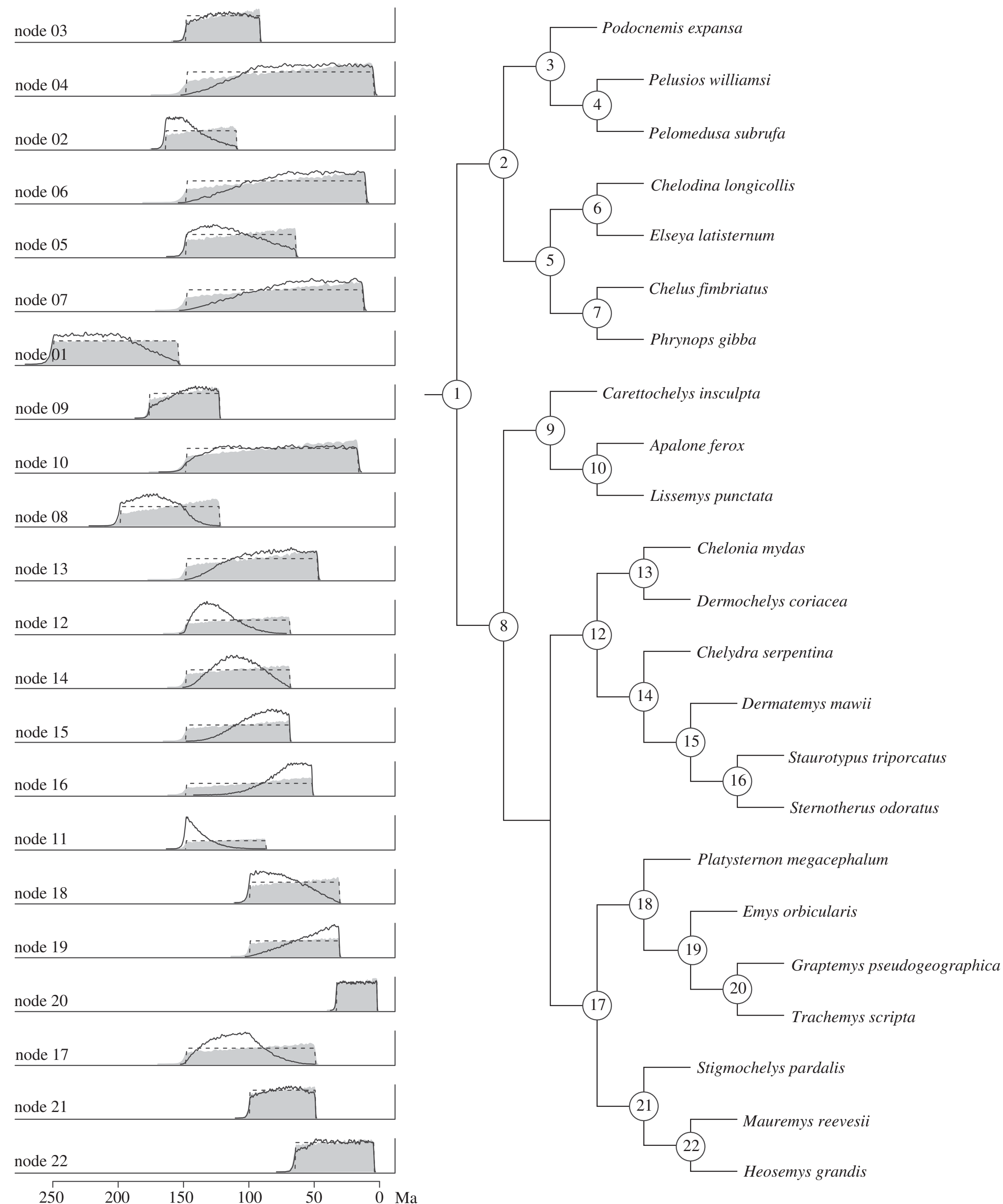
*“The [soft] maximum constraint is established as older than all the oldest possible records, extending back to encompass a time when the ecologic, biogeographic, geologic, and taphonomic **conditions for the existence of the lineage are met, but no records are known.**”*

The user “specified priors” will not (always) match the “effective priors” used during analysis



We can run our analysis “under the prior” (i.e., ignoring the sequence data)
to see how the software actually constructs the prior density
— accounts for the interaction between nodes

3.



An example from turtles 

Dashed lines = the specified priors

Grey shaded area = effective priors for one calibration, analysed alone

Black shaded area = effective priors for all calibrations together

The model doesn't describe the process that generated the fossil sampling times, meaning the model is **statistically incoherent**

The calibration priors are difficult to specify objectively and can have a massive impact on the divergence times. They can also interact with each other and / or the birth-death process prior in unintuitive ways

Some references on issues with specified vs effective priors

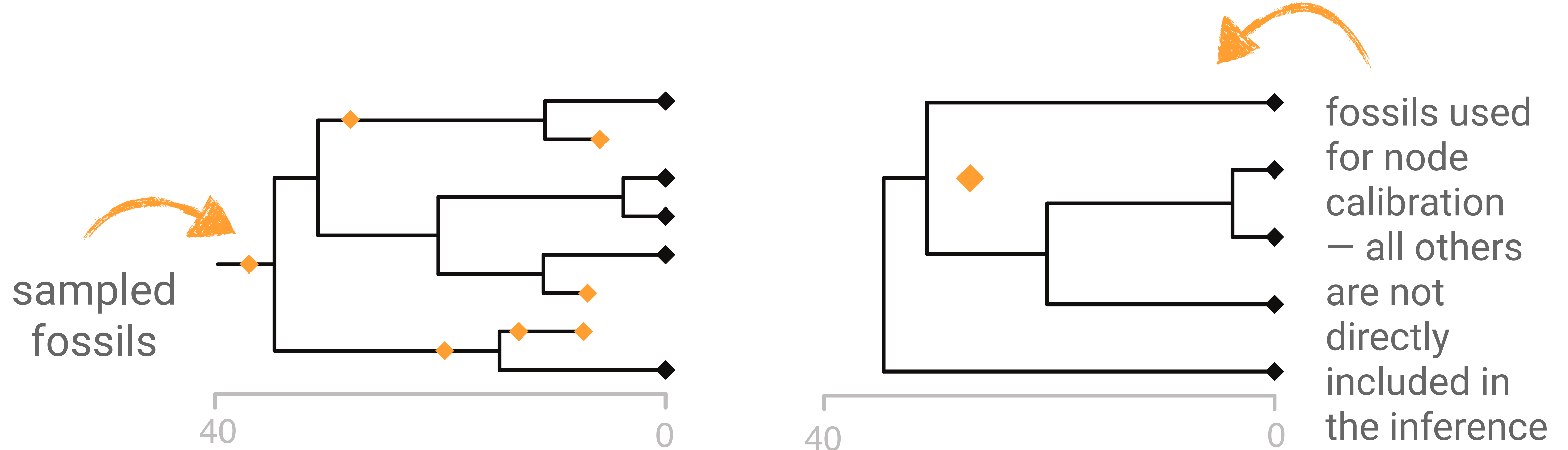
Yang and Rannala ([2006](#))

Heled and Drummond ([2012](#))

Warnock et al. ([2012](#), [2015](#))

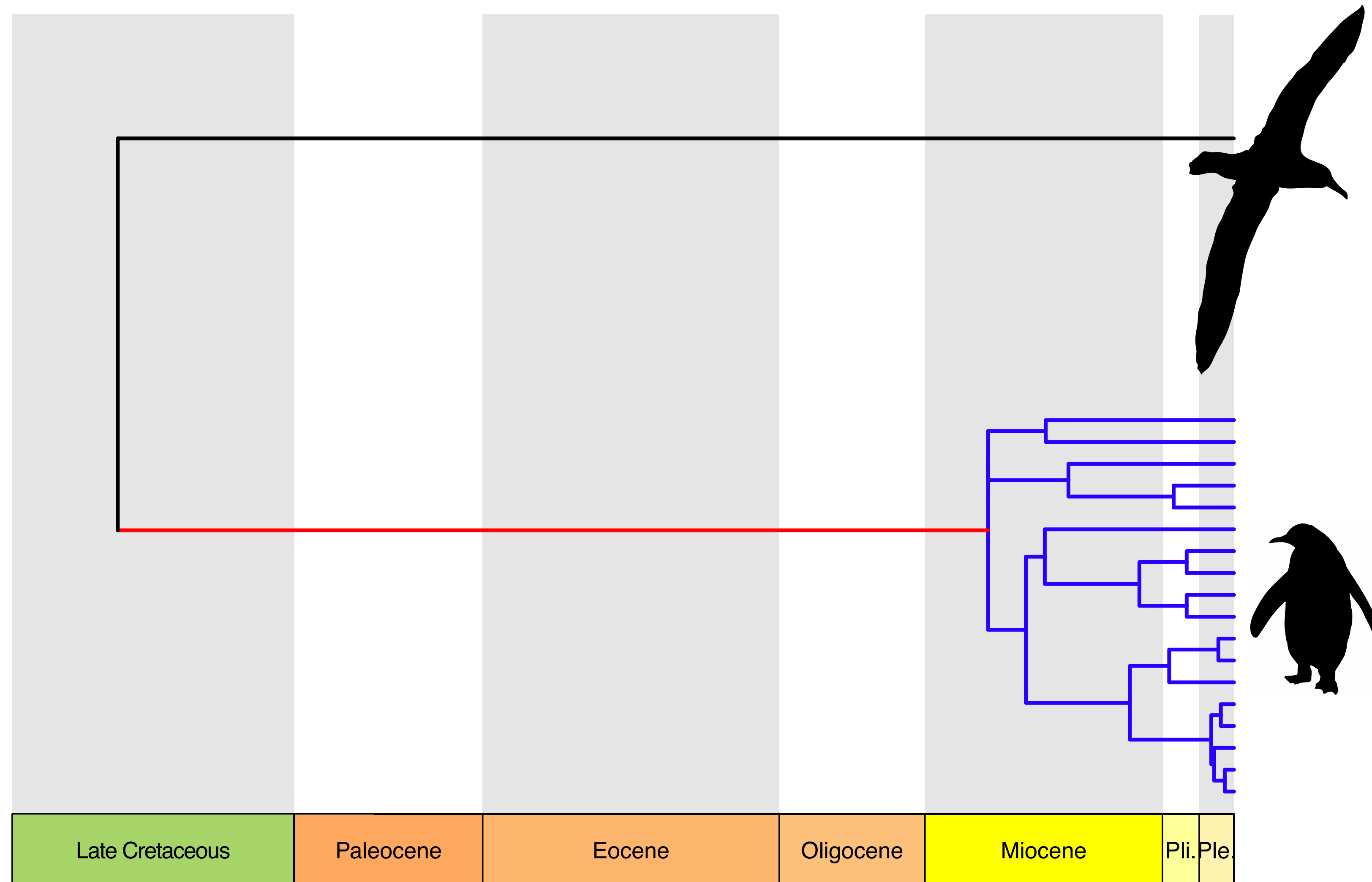
Not all data is included during inference

A lot of information is not directly included during inference, since typically we assign one fossil per calibration node



(although in theory all fossils could be used to construct probabilistic priors calibrations)

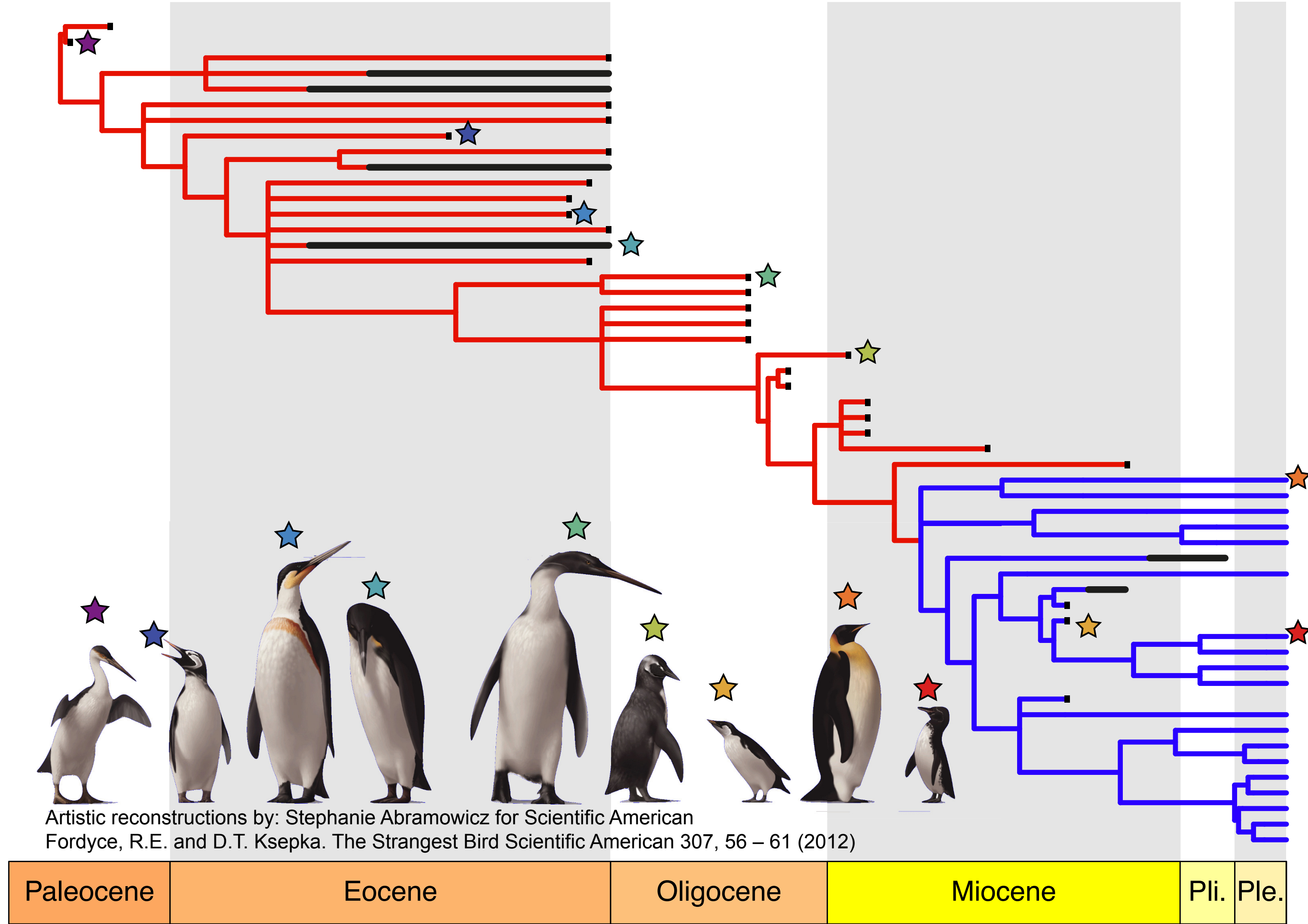
Example: living penguins



Nearest living relative is the group containing albatrosses and other sea birds - separated by ~60 Ma

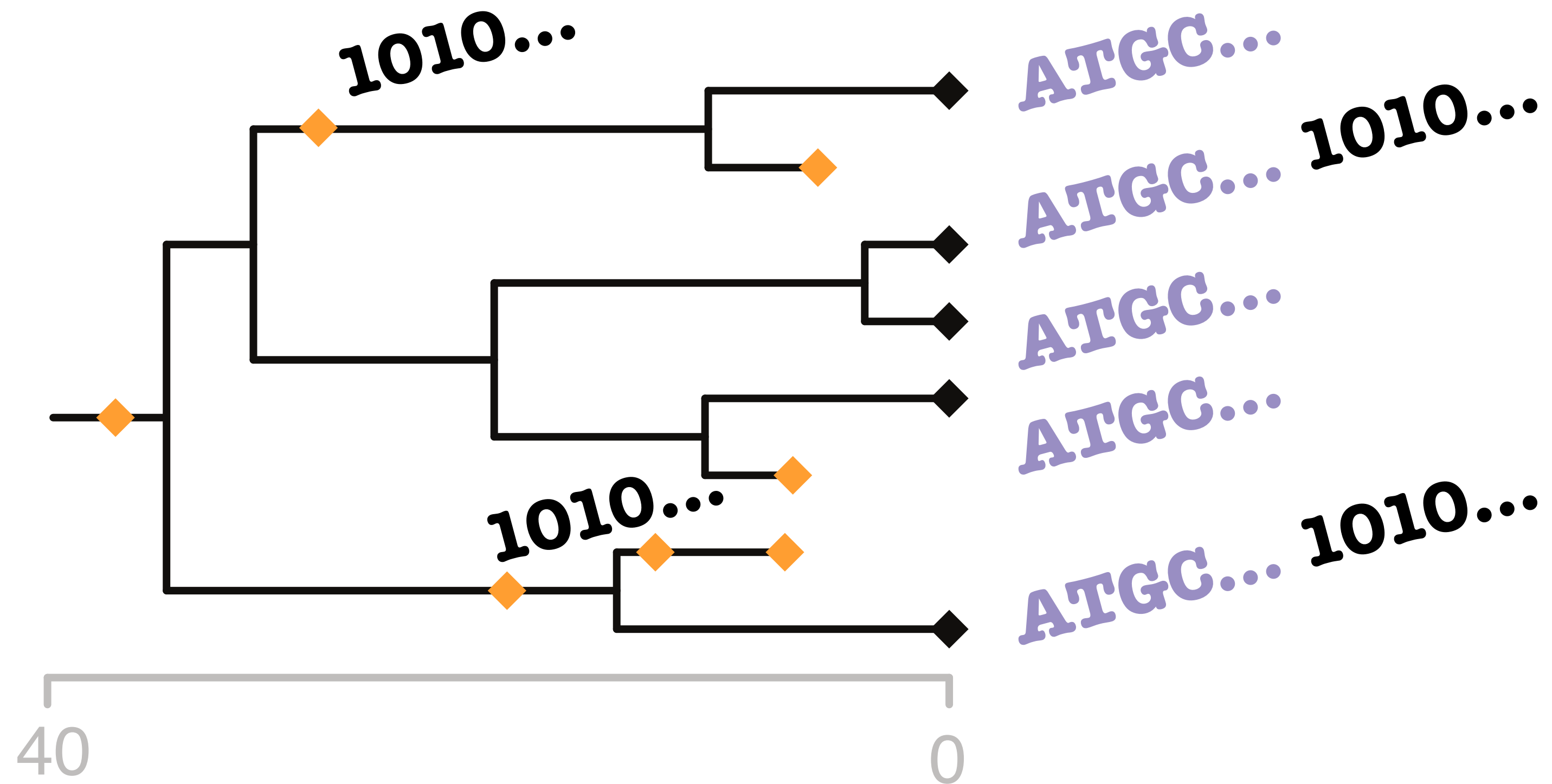
Example: living penguins

But penguins have a rich fossil record spanning the Cenozoic



Tip-dating / total-evidence dating

Tip-dating or “total-evidence” dating



We have DNA for living species. We have morphology for living *and* fossil species

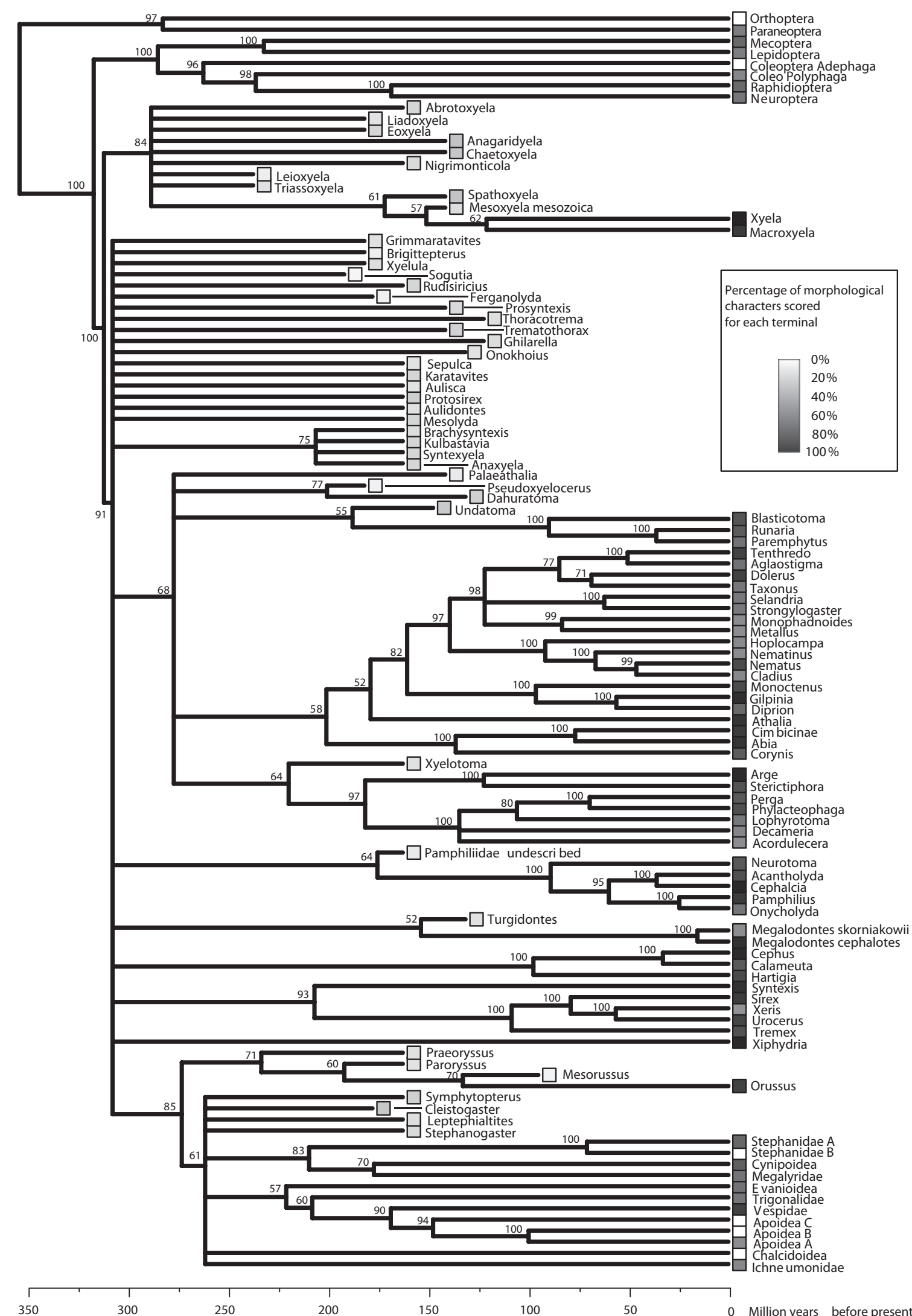
The phylogenetic position of fossils can be guided by morphology
→ accounts for uncertainty in fossil placement

The uniform tree prior

The uniform tree prior assumes all trees and branch lengths are equally likely within the bounds of the fossil ages (+ a max upper bound)



It does not explicitly account for the fossil sampling process



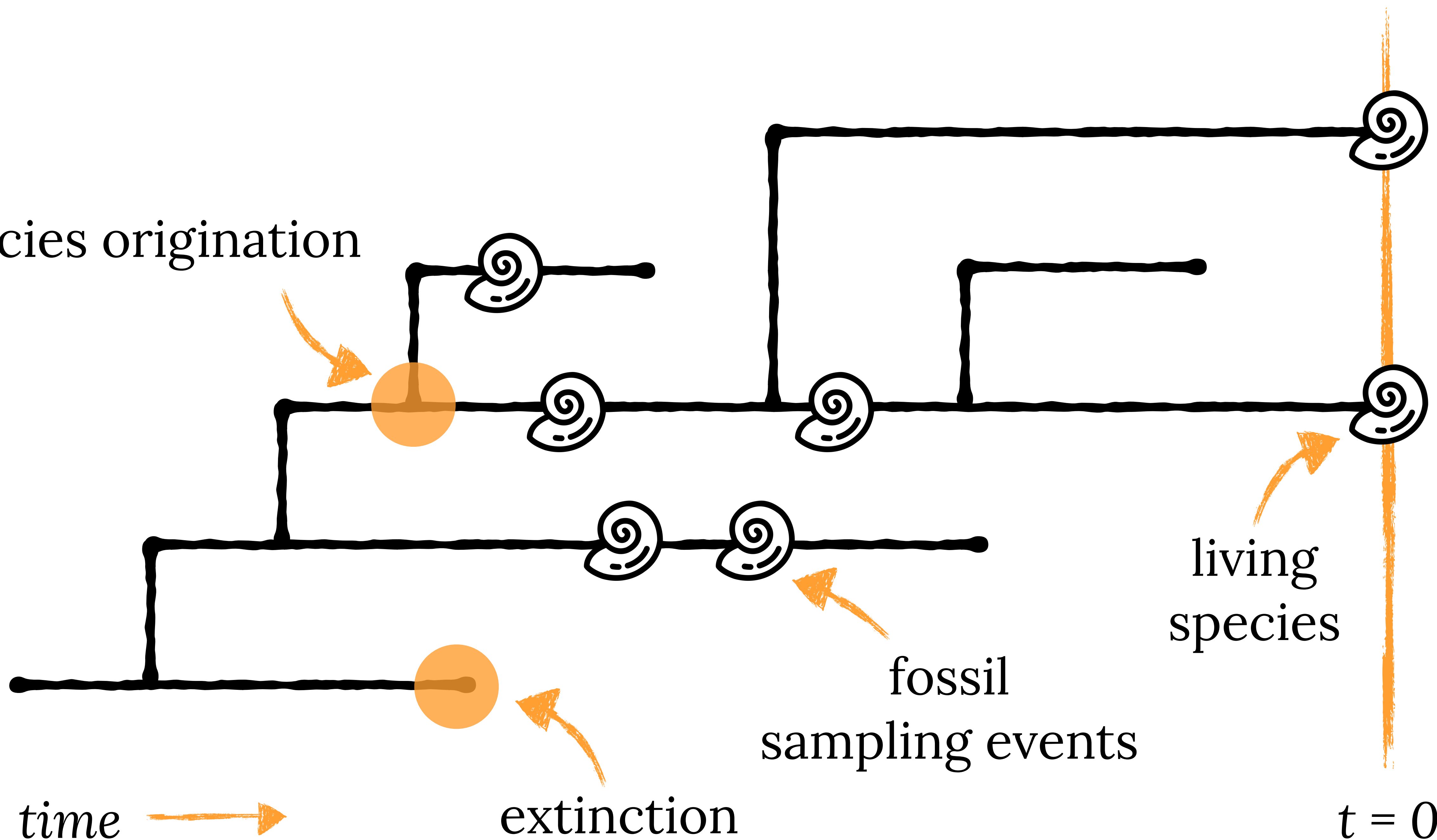
Dated tree of Hymenoptera

Ronquist et al. (2012) Sys Bio
Problems with uniform tree priors: see May et al. (2021)
[Uniform tree priors - why not use them?](#) Remco Bouckaert

What does a generating process for a tree containing fossils look like?

The fossilised birth-death process

species origination



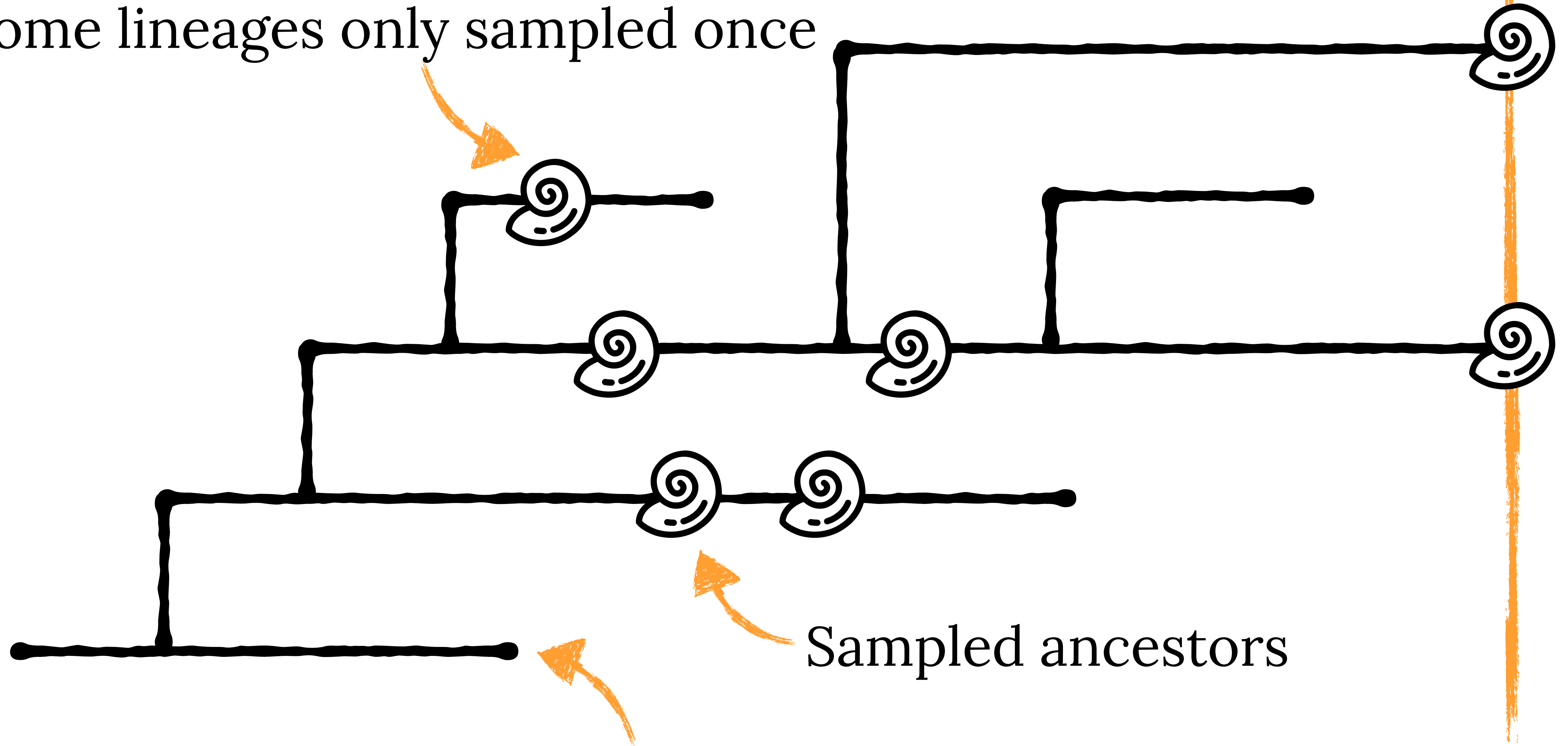
living
species

fossil
sampling events

extinction

$t = 0$
29

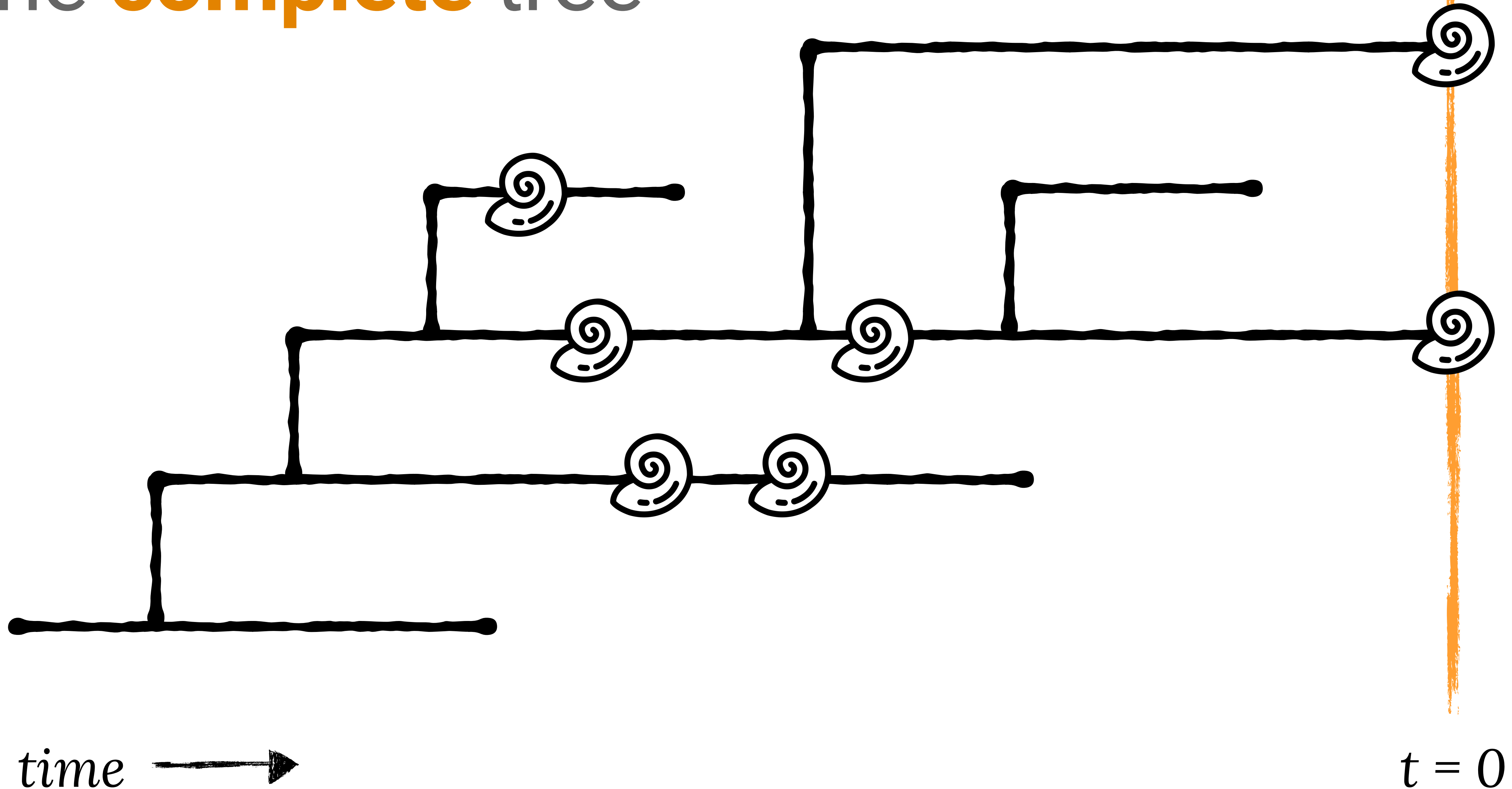
Some lineages only sampled once



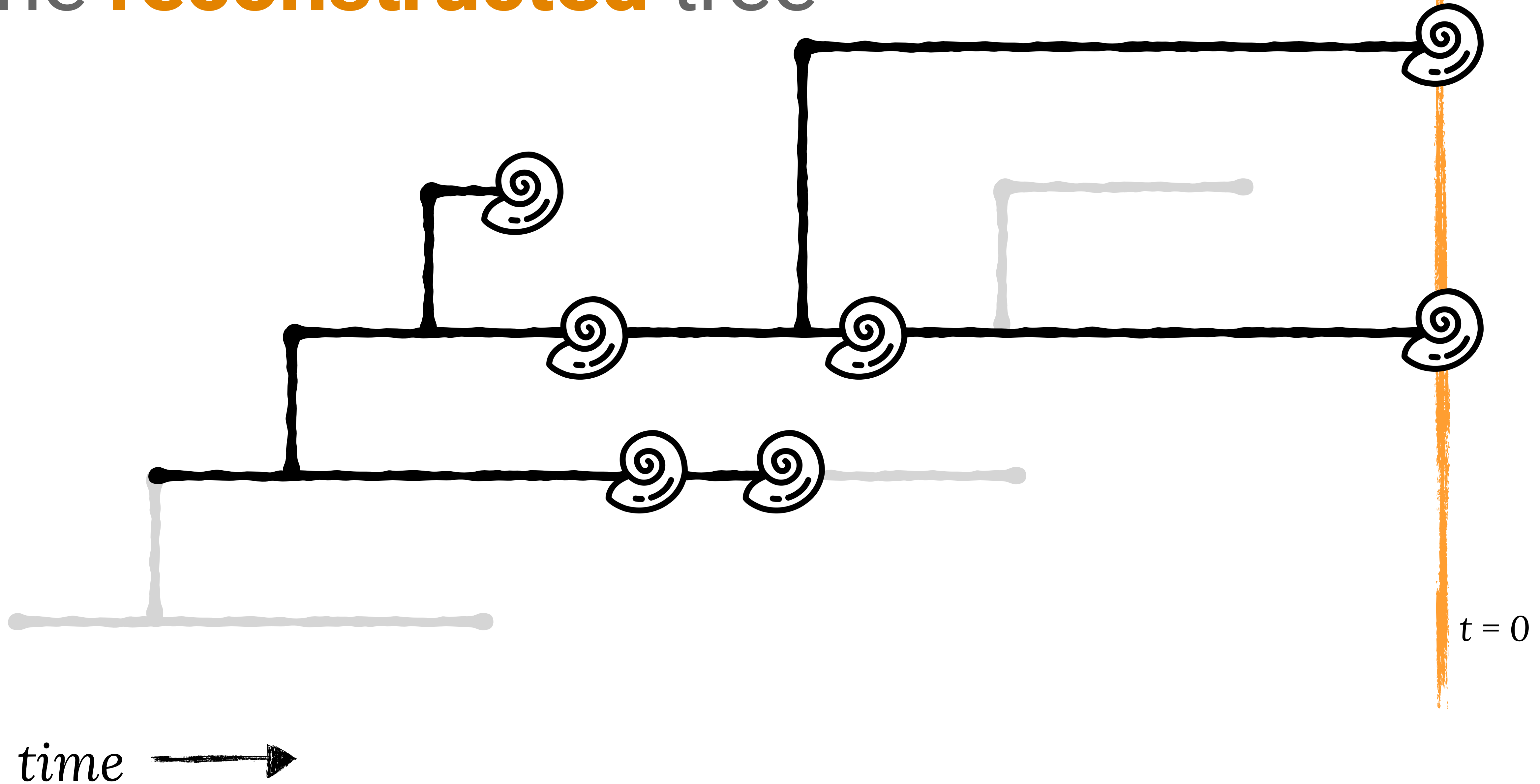
Some lineages go completely unsampled

$t = 0$

The complete tree


$$t = 0_{31}$$

The reconstructed tree



The **fossilised birth-death (FBD) process** allows us to calculate the probability of observing the reconstructed tree



$$P(\mathcal{E} | \text{spiral}, \lambda, \mu, \psi, \rho)$$



Sampling-through-time in birth-death trees. Stadler. ([2010](#))
First implemented: Heath et al. ([2014](#)) and Gavryushkina et al. ([2014](#))

Why is the FBD process so important?

There are many statistical advantages to having a generative model for time tree inference:

- better reflection of uncertainty
- increased flexibility
- greater accuracy

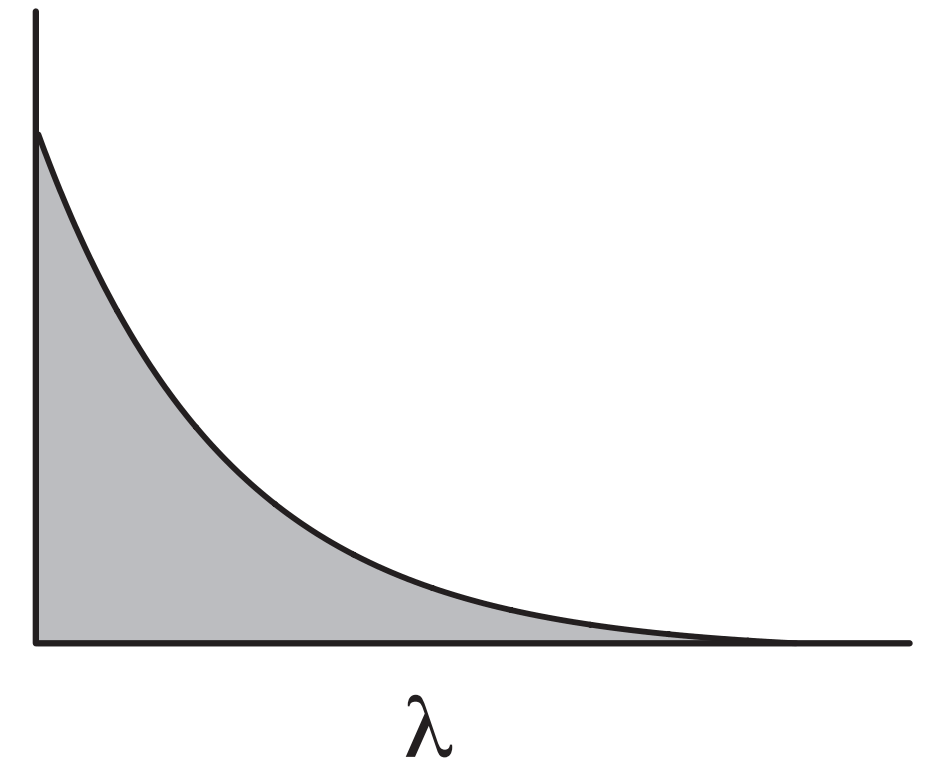
We can include fossils directly during inference

We can combine data in ways that weren't previously possible, infer diversification rates (phylodynamics), and even link the model parameters to abiotic processes

Key model parameters

- birth (speciation) rate — λ
- death (speciation) rate — μ
- fossil sampling rate — ψ
- extant sampling probability — ρ

birth, death and sampling are instantaneous rates and typically estimated during inference



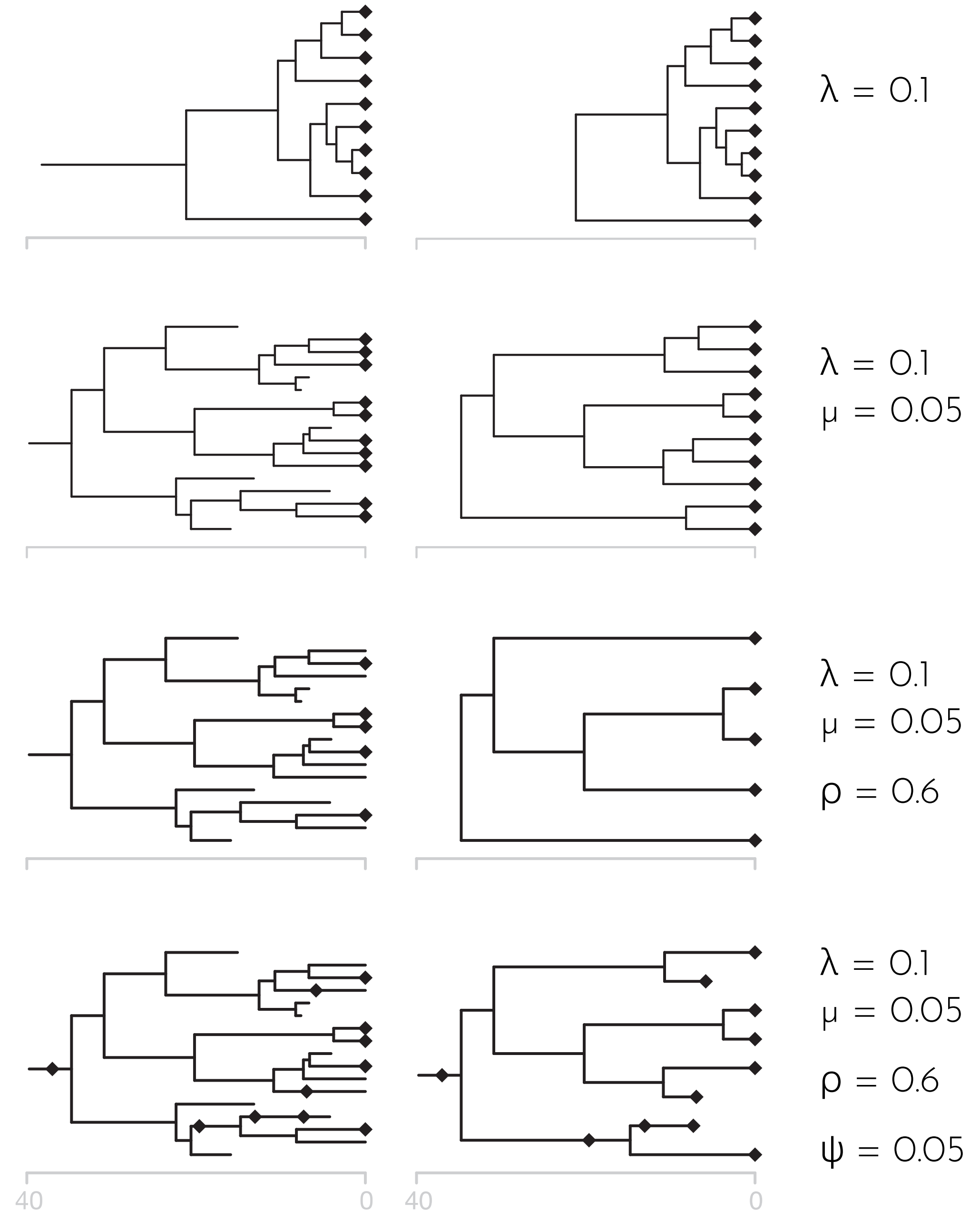
extant species sampling is often fixed because this is the parameter we have the most information about

Relationship to (some) other birth-death process models

We can also use ρ at $t > 0$ to model serial sampling

For epidemiology: Stadler et al. ([2012](#))
 See also: Stadler and Yang ([2013](#))
 Review: MacPherson et al. ([2022](#))
 Image: Warnock and Wright ([2020](#))

complete vs. reconstructed trees



Yang and
Rannala
([1997](#))
Stadler
([2009](#))

Stadler
([2010](#))

Parameterisation of the process

We can put priors on different combinations of parameters

Parameter	Transformation
Net diversification	$d = \lambda - \mu$
Turnover	$v = \mu/\lambda$
Sampling proportion	$s = \psi/(\mu + \psi)$
Speciation	$\lambda = d/(1 - v)$
Extinction	$\mu = (vd)(1 - v)$
Sampling	$\psi = (s/(1 - s))((vd)/(1 - v))$

Parameterisation of the process

For **epidemiology** we typically condition on the sampling proportion, and exclude extant species sampling. We are ultimately interested in a different set of parameters

We can also introduce r , the probability of being removed from the process at sampling

Parameter	Transformation
Leaf sampling proportion	$s_l = \psi r / (\mu + \psi r)$
Total removal rate	$\delta = \mu + \psi r$
Effective reproductive number	$R_0 = \lambda / (\mu + \psi r)$

Fossils can be incorporated via **taxonomy** or **character data** (total-evidence)

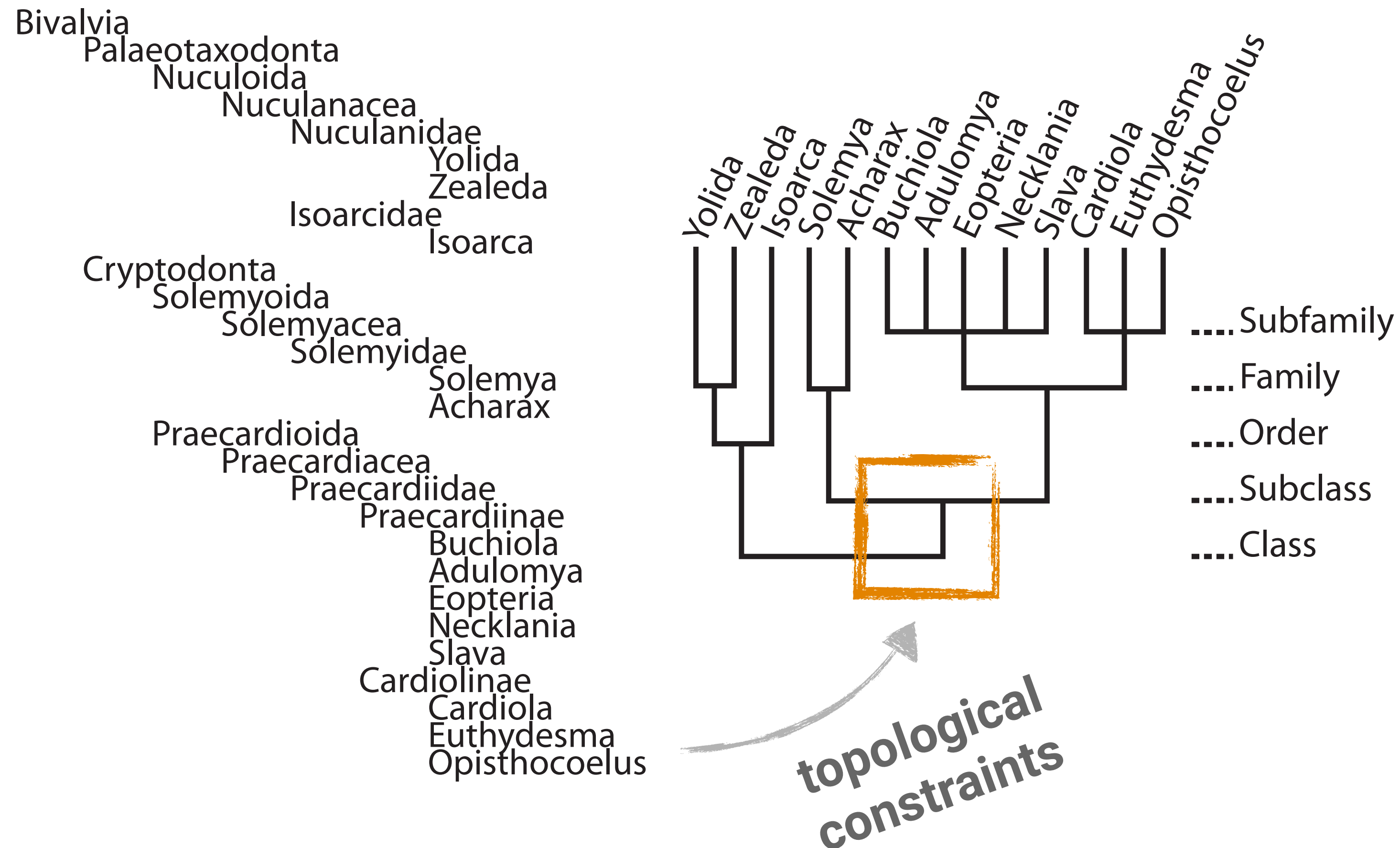
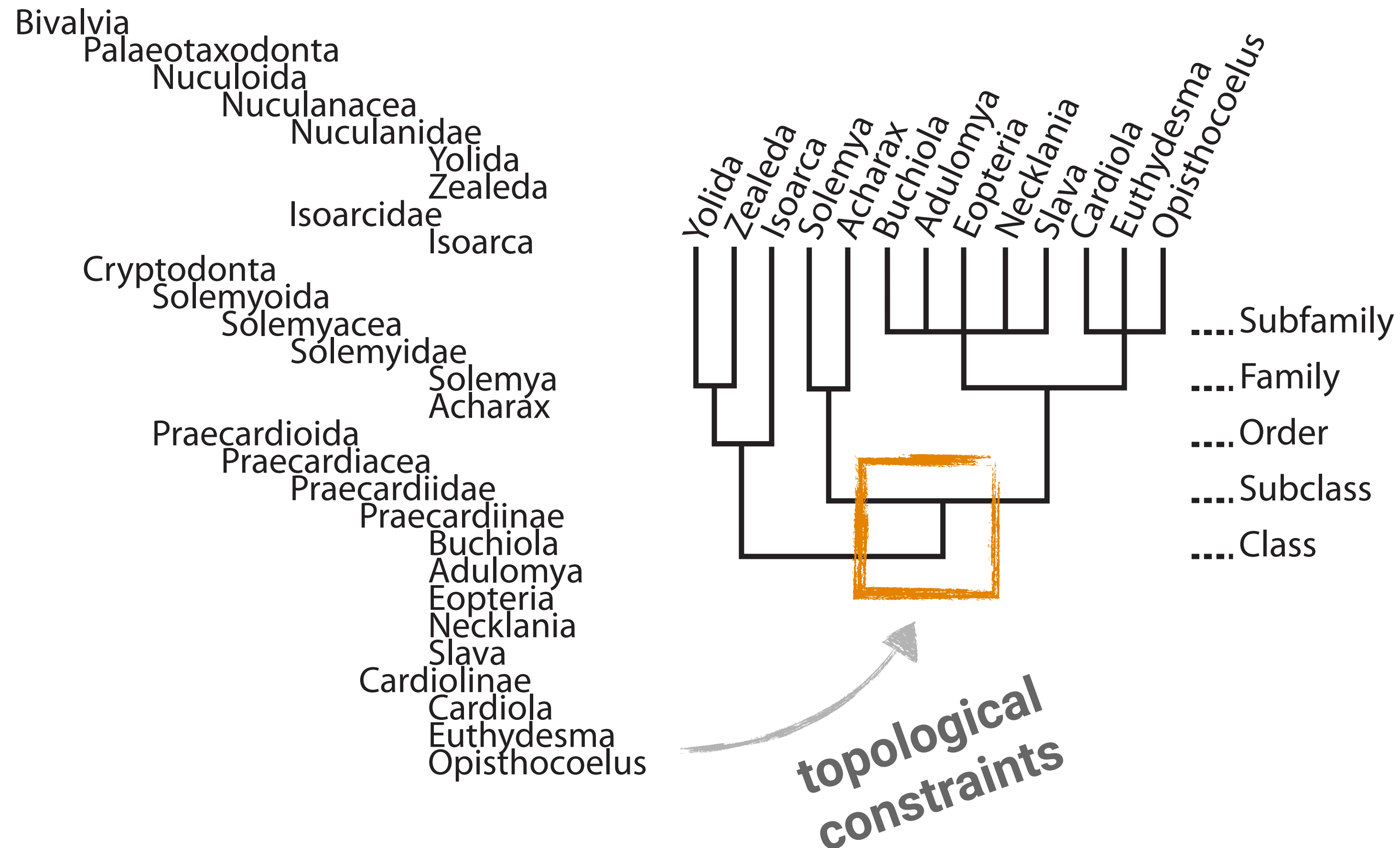


Image source Soul & Friedman ([2015](#))

Fossils can be incorporated via **taxonomy** or **character data** (total-evidence)



A T A T ...

TCAC...

????...

OR

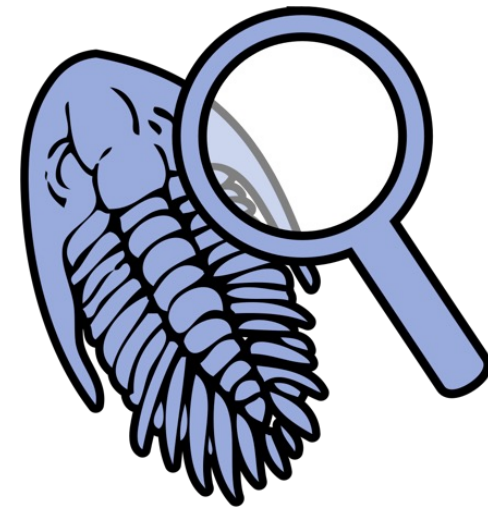
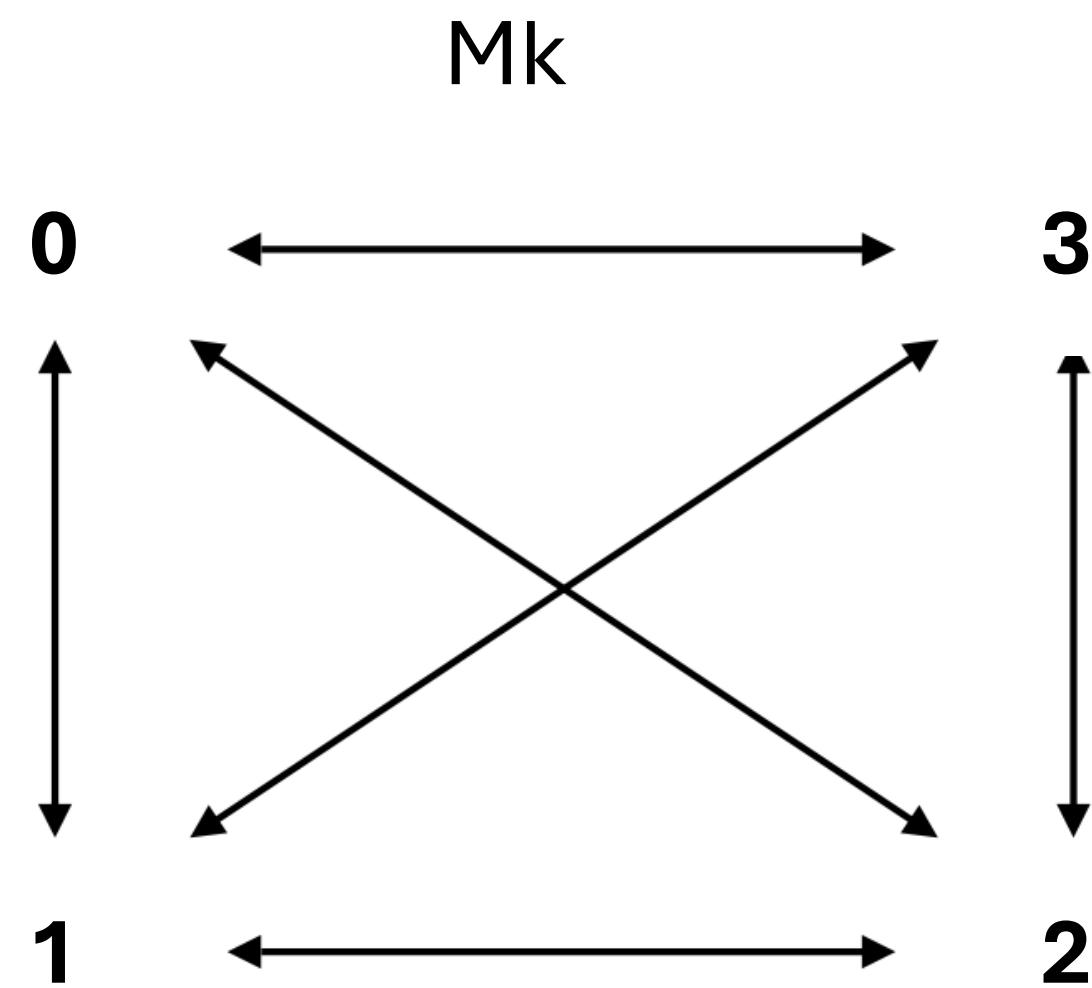
1001...

1101...

0100...

Image source Soul & Friedman ([2015](#))

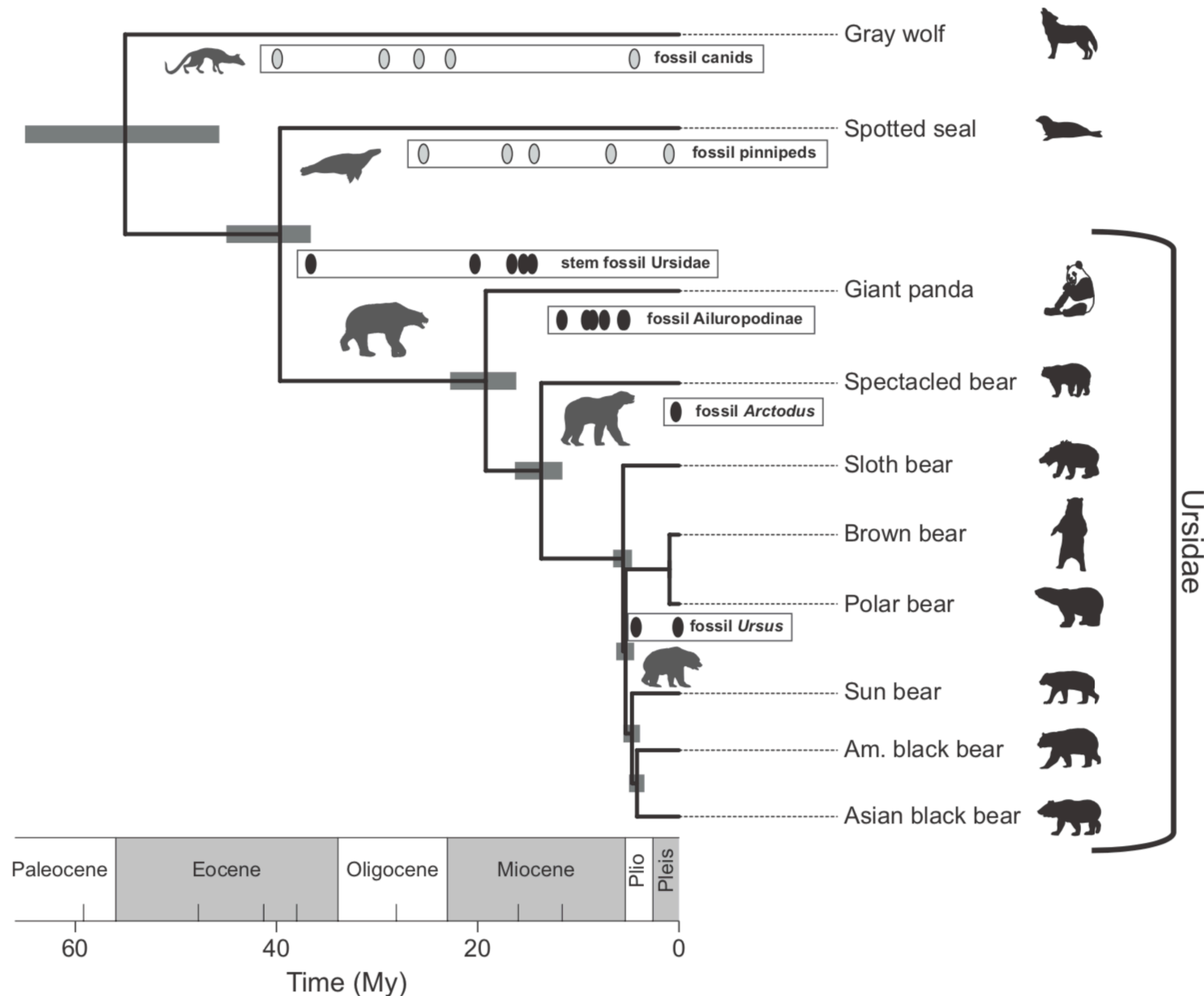
Mk model



$$Q = \begin{pmatrix} -\mu_0 & \mu_{01} & \mu_{02} & \mu_{03} \\ \mu_{10} & -\mu_1 & \mu_{12} & \mu_{13} \\ \mu_{20} & \mu_{21} & -\mu_2 & \mu_{23} \\ \mu_{30} & \mu_{31} & \mu_{32} & -\mu_3 \end{pmatrix},$$

*4 state here as an example, can be any number from 2!

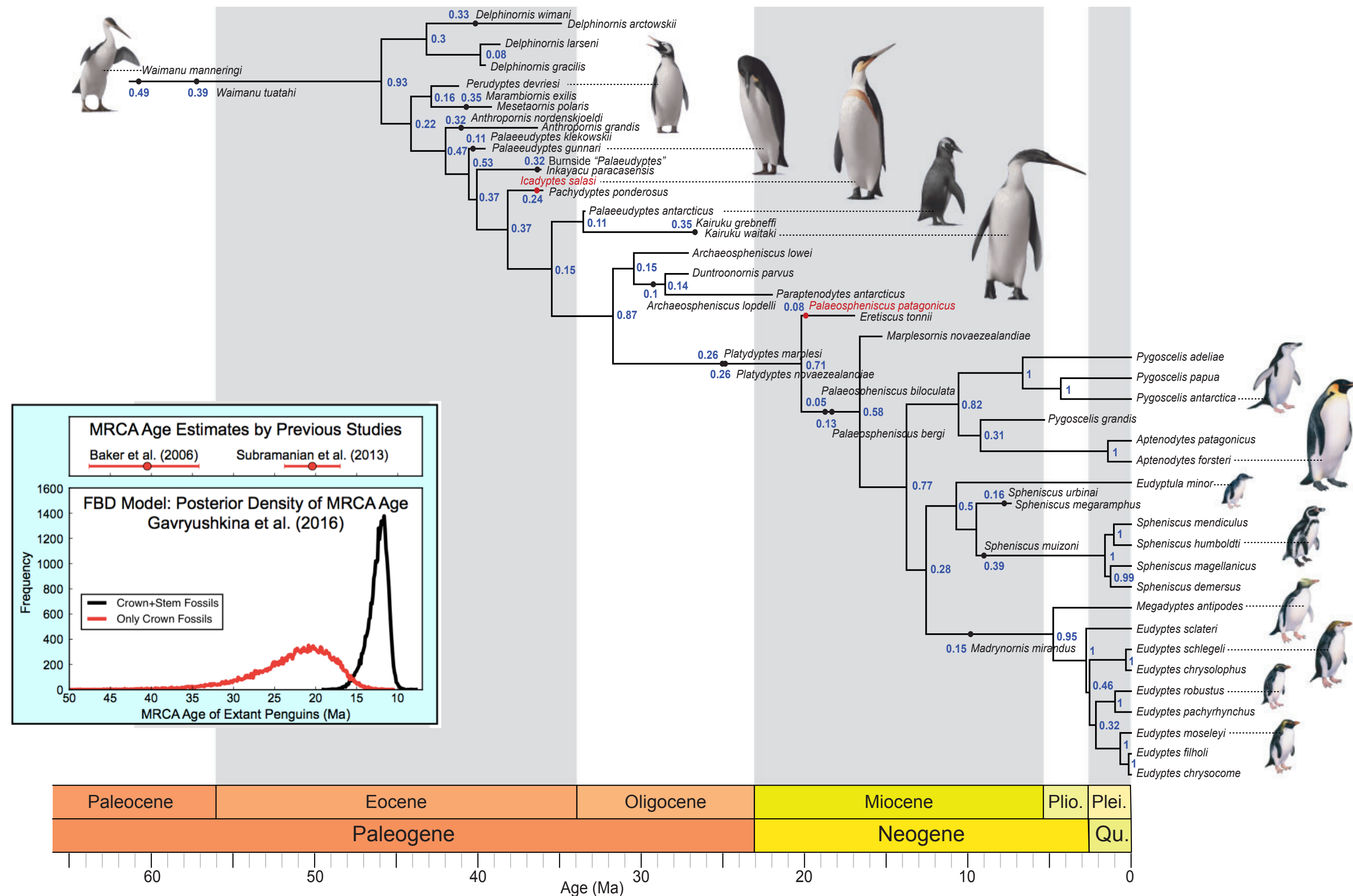
Time calibrated tree of living and fossil bears



First application of the FBD model

Fossils are incorporated via constraints, **not character data**. Their precise placement can not be inferred, but this uncertainty will be reflected in the posterior

Time calibrated tree of living and fossil penguins



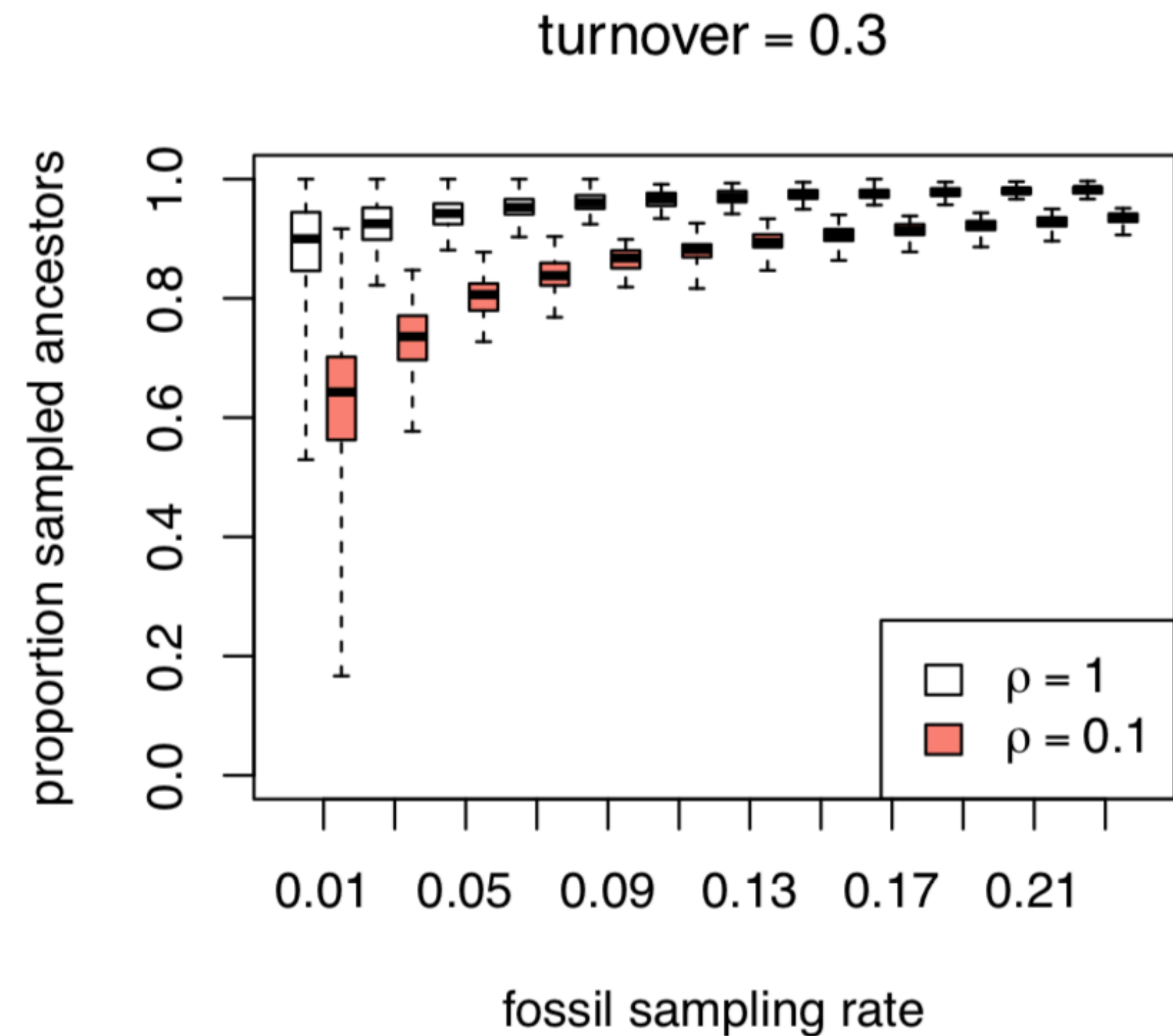
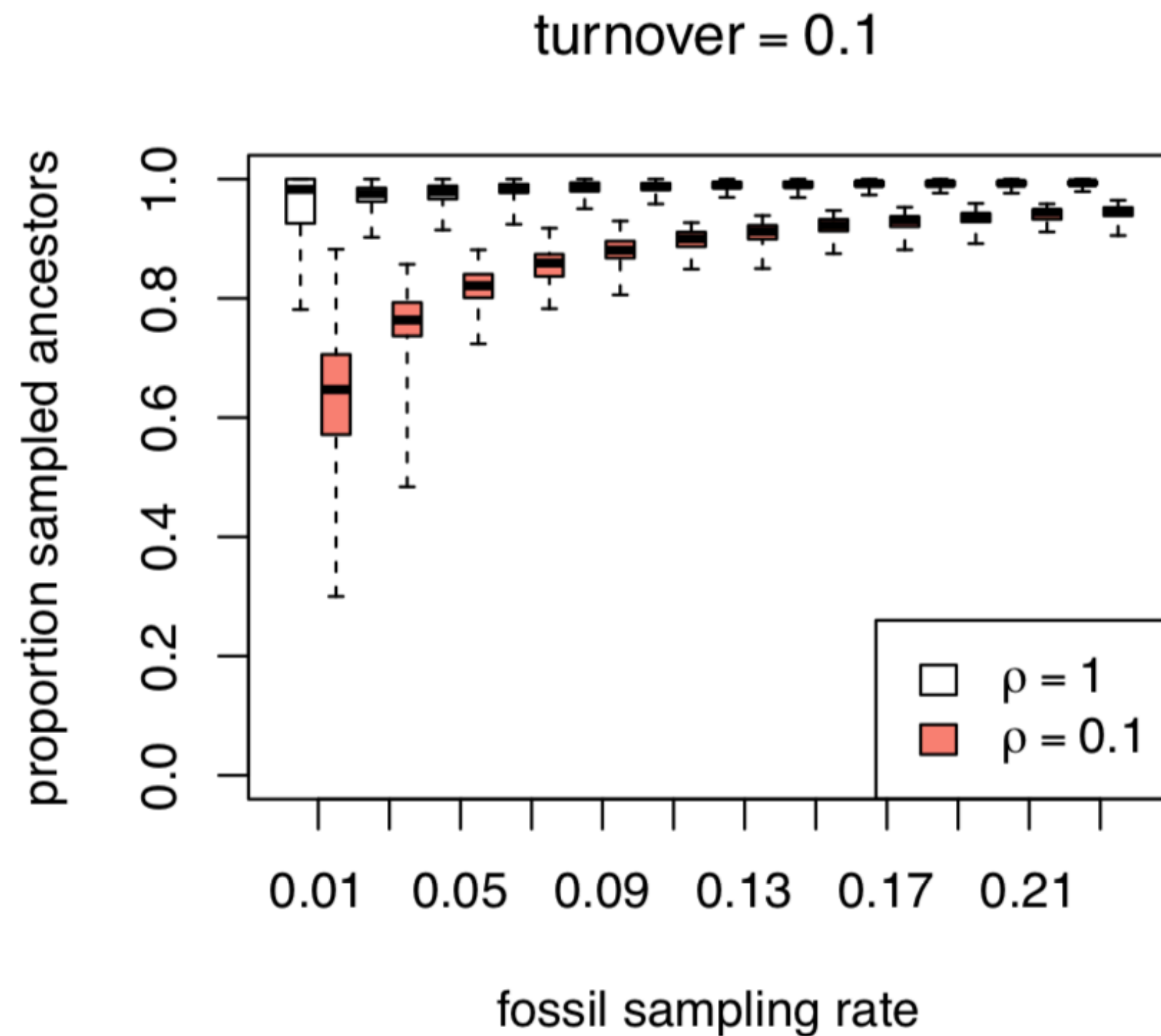
First application of total-evidence dating using the FBD model

Fossils are incorporated using character data

Gavryushkina et al. ([2016](#))
See also Zhang et al. ([2016](#))

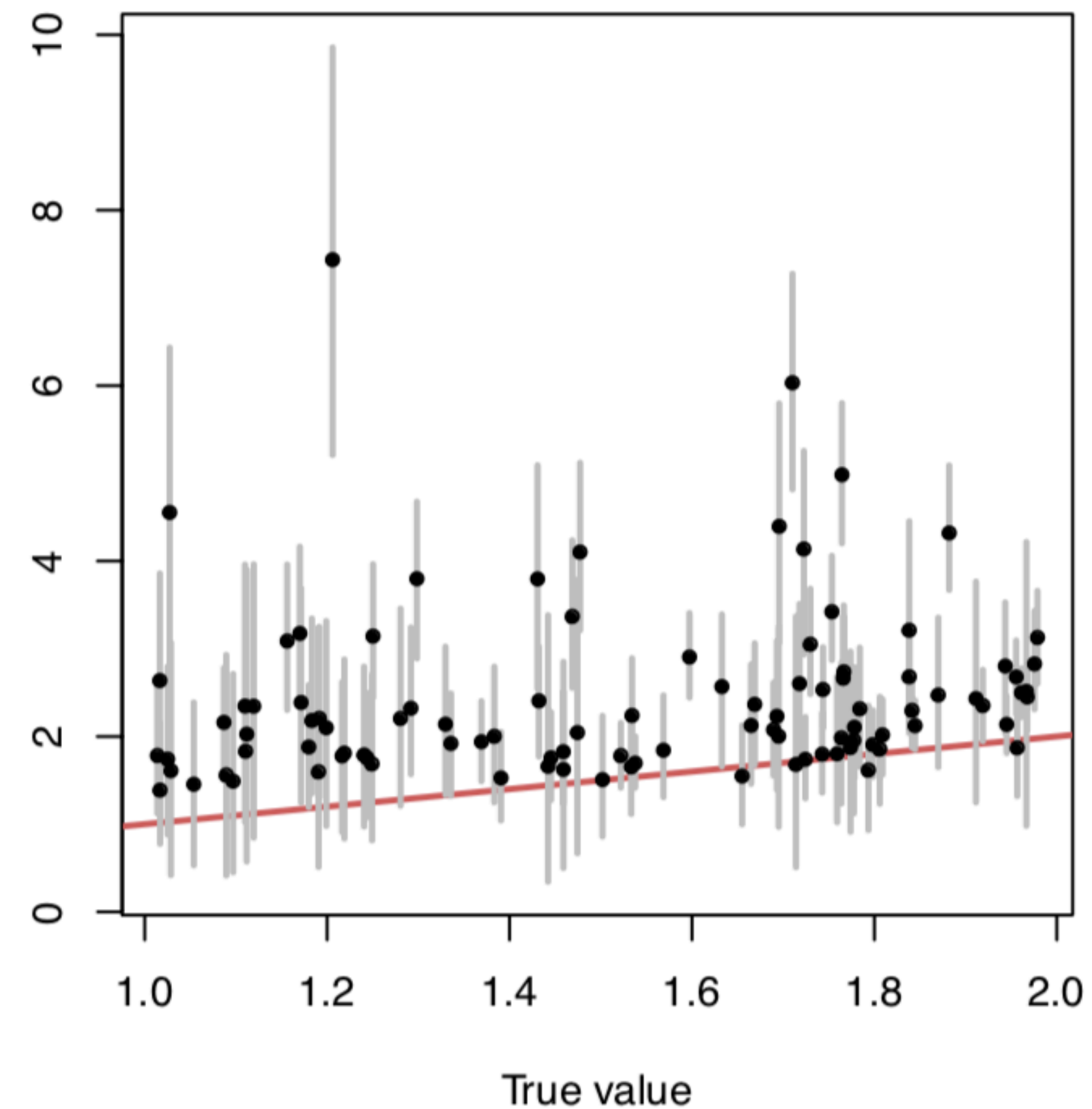
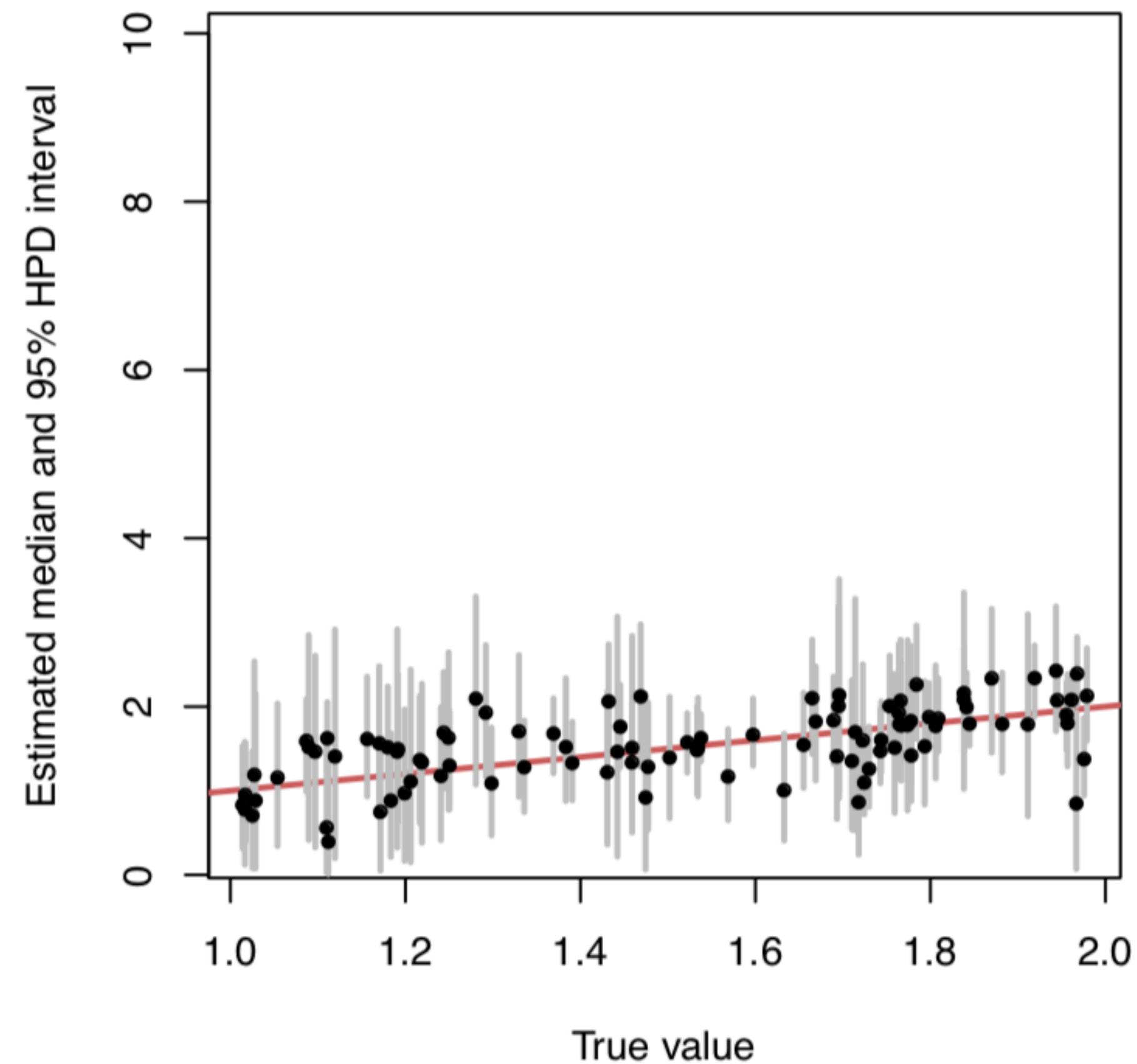
Sampled ancestors

The proportion increases with higher turnover (birth - death) or higher sampling

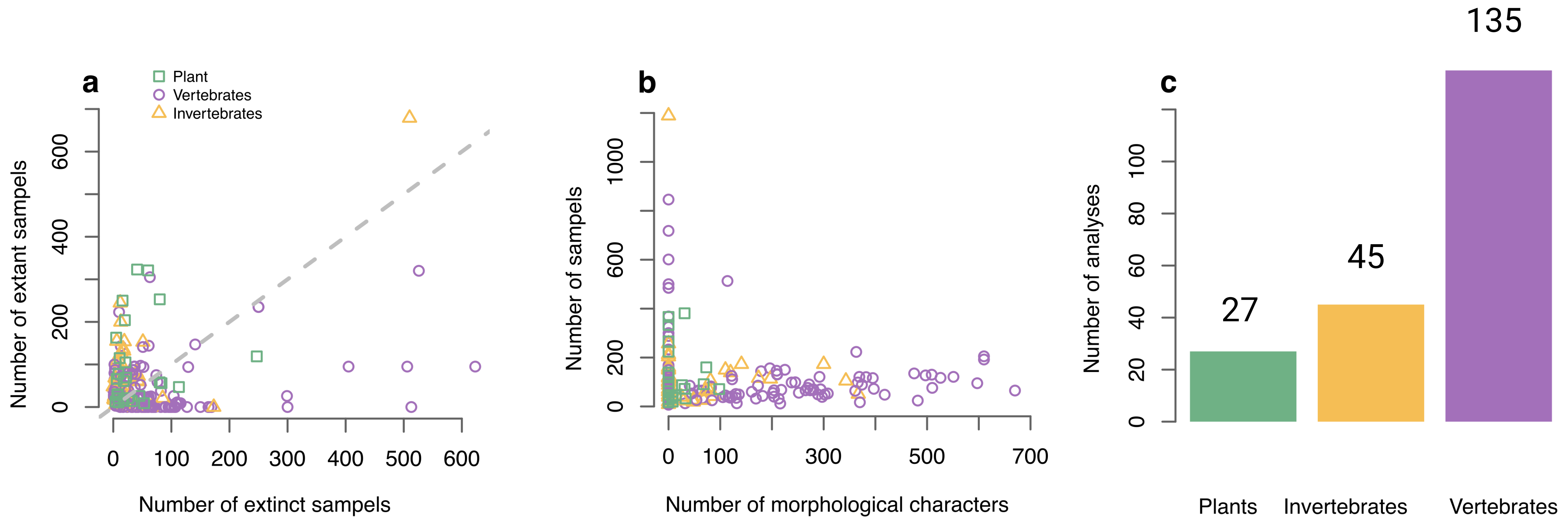


Sampled ancestors

Ignoring sampled ancestors can lead to inaccurate parameter estimates

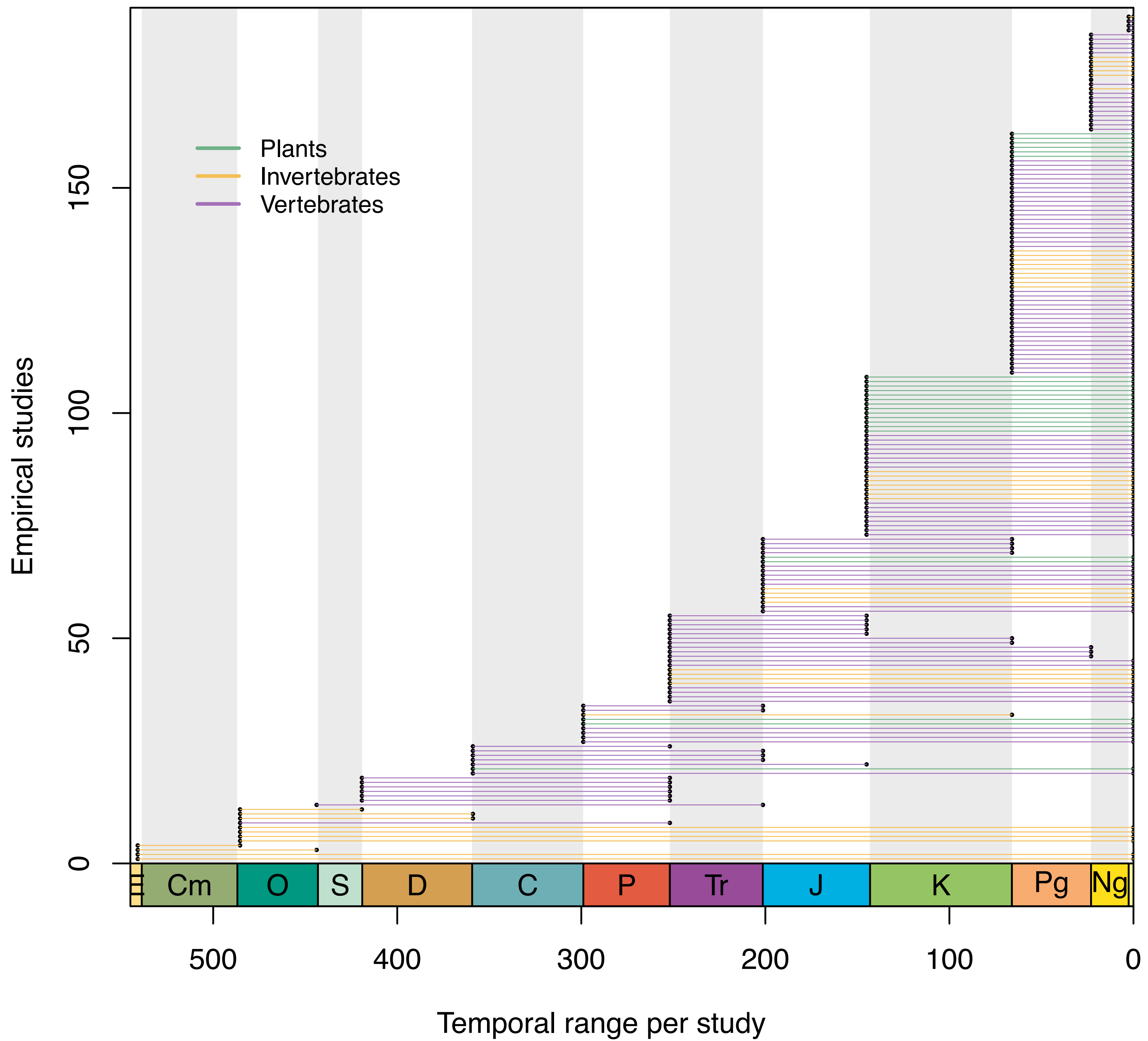


Applications of the FBD model to date



Literature survey: FBD use to date

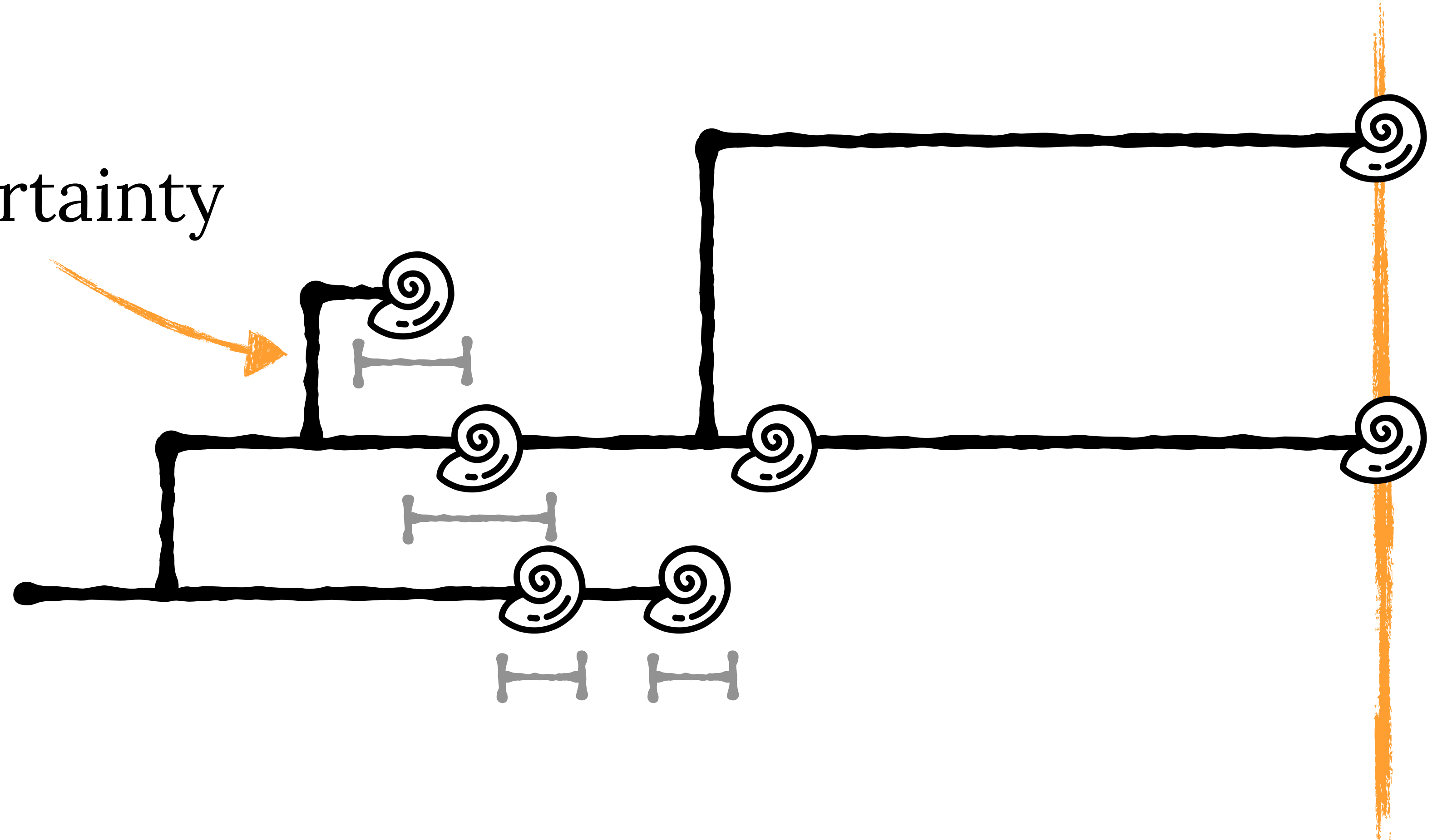
- 176 studies, with 208 empirical analyses, applying the FBD process (until February 2024)
- Used across a huge range of time intervals, data sets (mean extant samples = 74, mean extinct samples = 60)
- 109 studies used BEAST2, 93 used MrBayes, 10 used RevBayes



Analysis type	Phylogenetic data			No. of analyses
	Molecular	Morphology	Morphology [†]	
Total evidence	✓	✓	✓	53
Extant only	✓			78
Morphology		✓	✓	26
Extinct only			✓	35
No phylogenetic data				16

Sample age uncertainty

age uncertainty



Barido-Sottani et al. ([2018](#), [2020](#)), Ignoring fossil age uncertainty leads to inaccurate topology in time calibrated tree inference

*Barido-Sottani et al. ([2023](#)) Putting the F in FBD analyses: tree constraints or morphological data? *Palaeontology* 50*

FBD analyses
need taxonomists!

OCTOPI WALL STREET

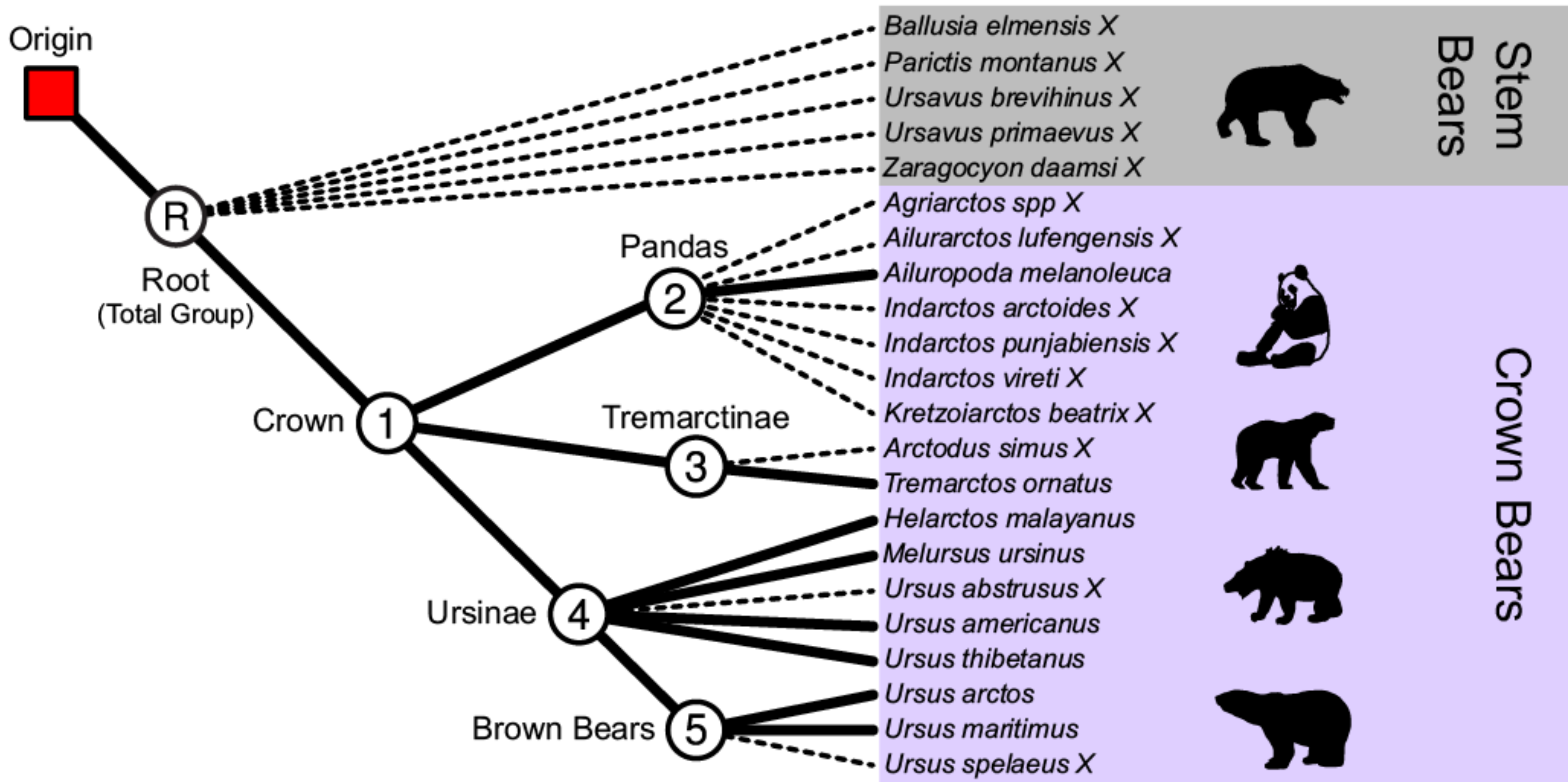


Invertebrates are 97% of animal diversity!

Brought to you by Oregon Institute of Marine Biology, University of Oregon

Exercise

Make sure you have the latest version of ORC package installed (v 1.2.1)



Data

2 genes (irbp, cytb) for 8 extant species

Occurrence times of 14 fossil species

Taxonomic constraints

Note: this analysis includes includes **no character data** for our fossils

More about models of morphological evolution / TE dating on Thu / Fri

←

→

Search

≡ bears_cytb_fossils.nex

≡ bears_irbp_fossils.nex ×

Users > warnock > Desktop > org > 02_events > TtB > FBD_tutorial > ≡ bears_irbp_fossils.nex

1 #NEXUS

2

3 begin data;

4 dimensions ntax=22 nchar=1281;

5 format datatype=dna missing=? gap=-;

6 matrix

7 Ailuropoda_melanoleuca_0 CCAGAGAACCTGATGGGGATGCAGGAAGCCATCGAGCAGGCCACCAAGAGTCGTGAGATCCTCGCCATCTCAGACCCTCAGACTCTGGCCCATGTGCTGACCACTGGGGTGCA

8 Tremarctos_ornatus_0 CCAGAGAACCTGATGGGGATGCAGGAAGCCATCGAGCAGGCCACCAAGAGTCGTGAGATCCTCGCCATCTCAGACCCTCAGACTCTGGCCCATGTGCTGACCACTGGGGTGCA

9 Helarctos_malayanus_0 CCAGAGAACCTGATGGGGATGCAGGAAGCCATCGAGCAGGCCACCAAGAGTCGTGAGATCCTCGCCATCTCAGACCCTCAGACTCTGGCCCATGTGCTGACCACTGGGGTGCA

10 Melursus_ursinus_0 CCAGAGAACCTGATGGGGATGCAGGAAGCCATCGAGCAGGCCACCAAGAGTCGTGAGATCCTCGCCATCTCAGACCCTCAGACTCTGGCCCATGTGCTGACCACTGGGGTGCA

11 Ursus_americanus_0 CCAGAGAACCTGATGGGGATGCAGGAAGCCATCGAGCAGGCCACCAAGAGTCGTGAGATCCTCGCCATCTCAGACCCTCAGACTCTGGCCCATGTGCTGACCACTGGGGTGCA

12 Ursus_arctos_0 CCAGAGAACCTGATGGGGATGCAGGAAGCCATCGAGCAGGCCACCAAGAGTCGTGAGATCCTCGCCATCTCAGACCCTCAGACTCTGGCCCATGTGCTGACCACTGGGGTGCA

13 Ursus_thibetanus_0 CCAGAGAACCTGATGGGGATGCAGGAAGCCATCGAGCAGGCCACCAAGAGTCGTGAGATCCTCGCCATCTCAGACCCTCAGACTCTGGCCCATGTGCTGACCACTGGGGTGCA

14 Ursus_maritimus_0 CCAGAGAACCTGATGGGGATGCAGGAAGCCATCGAGCAGGCCACCAAGAGTCGTGAGATCCTCGCCATCTCAGACCCTCAGACTCTGGCCCATGTGCTGACCACTGGGGTGCA

15 Parictis_montanus_36.6 ??

16 Zaragocyon_daamsi_21.9 ??

17 Ballusia_elmensis_14.0 ??

18 Ursavus_primaevus_14.4 ??

19 Ursavus_brevihinus_16.2 ??

20 Indarctos_vireti_8.7 ??

21 Indarctos_arctoides_9.6 ??

22 Indarctos_punjabiensis_5.0 ??

23 Ailurarctos_lufengensis_7.7 ??

24 Agriarctos_spp_5.0 ??

25 Kretzoiarctos_beatrix_11.7 ??

26 Ursus_abstrusus_4.3 ??

27 Ursus_spelaeus_0.06 ??

28 Arctodus_simus_0.5 ??

29 ;

30 end;

31

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Plain Text

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Take care when interpreting PP for
trees with SAs

More options available via the pdf!

e.g., alternative parameterisations, fully extinct clades